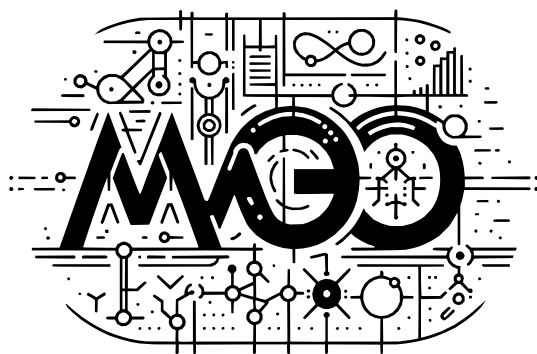


BOGDAN OKREŠA ĐURIĆ

DEVELOPING A FRAMEWORK FOR AGENT GAMIFICATION BASED ON ONTOLOGIES

M A G O

PHASE 2: MAGO-GA ONTOLOGY AND FRAMEWORK



UNIZG FACULTY OF ORGANIZATION AND INFORMATICS
ARTIFICIAL INTELLIGENCE LABORATORY

This document contains some of the results and deliverables of the project ID **MOBODL-2023-08-5618** funded by the European Union and the Croatian Science Foundation.



**Funded by
the European Union**
NextGenerationEU



Authors: Bogdan Okreša Đurić.

Editor: Bogdan Okreša Đurić, [dokresa \[at\] foi.unizg.hr](mailto:dokresa[at]foi.unizg.hr).

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Version: June 2025

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Phase 2

Introduction

The development and engineering processes of the ontology described here follow the steps defined by the METHONTOLOGY ontology engineering methodology [1], [2]. This is the methodology of choice because it is very well defined, using clearly described steps to engineer the desired ontology. This and a dozen other ontology engineering methodologies were analysed more thoroughly in preparation for the author's doctoral thesis [2].

This chosen ontology engineering methodology is described as having a set number of steps, each described in detail by the authors of the methodology. In addition, steps related to the entire life cycle of an ontology are identified. An ontology life cycle is described by Iqbal et al. as '[...][a] set of stages through which the ontology moves during its life.' [3, p. 2997]

© see figure 1

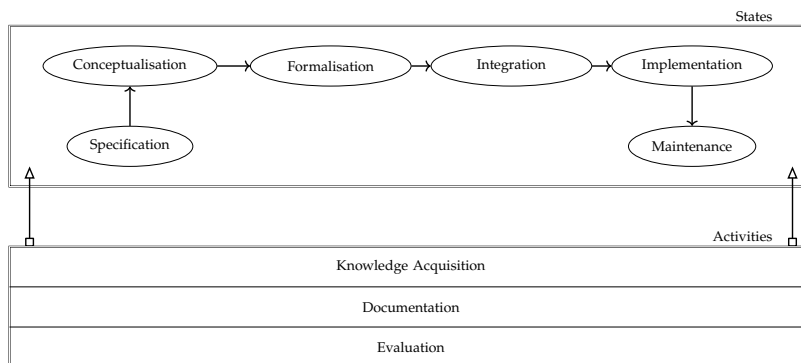


Figure 1: Basic steps of METHONTOLOGY ontology engineering methodology, reproduced from [2], adapted from [1]

The rest of this document is structured according to the defined METHONTOLOGY steps, representing the documentation process. First, the ontology specification document is described and given, followed by the description of knowledge acquisition as an ongoing process. Conceptualisation originates from the defined specification and is the input to formalisation. Once the ontology is formalised, it can be integrated with other ontologies of similar domains. Finally, the ontology is implemented and maintained. The ontology is evaluated in phase P1.2 of this part of this research. Each methodology step is briefly described at the beginning of the related chapter.

Part I

The Games as Intelligent Virtual Environments Ontology GIVEn

1

Specification

The second phase of this research is geared towards video games and the concepts related to video games. The main objective of this is to combine the concepts of cro:MASmultiagent systems (MASs) and video games into a working system capable of modelling video games using the concepts applicable to MASs, with special emphasis on modelling systems of agents in cro:IVEintelligent virtual environments (IVEs). Furthermore, the modelled MAGO-Ga ontology aims to map gamification's techniques, mechanics, actions and behaviours to propose a way of introducing gamification to a system of reasoning agents. The result will be developed towards a system that can automatically adapt based on the set or emerging goals while keeping true to its initially (or planned for) gamified nature.

The GIVEn ontology is the first step towards creating the MAGO-Ga ontology that features gamification-specific concepts.

The main objective of the GIVEn ontology is to provide concepts necessary for describing video games as intelligent virtual environments (IVEs). An IVE is a concept used for describing 'a physical world inhabited by autonomous intelligent entities.' [4], [5], [6] When transferred to the domain of video games, the concept of an IVE can be considered as describing the game world inhabited by autonomous intelligent entities. Furthermore, one might consider the 'physical world' concept to map to the 'game world', the people of the physical world as the artificial intelligent agents playing the cro:NPCnon-player characters (NPCs) in the game world, the 'autonomous intelligent entities' as additional artificial agents in the game world implementing all the other needed roles.

The GIVEn ontology will be used to model video games in the domain context of IVEs. Therefore, the GIVEn ontology builds upon and extends the already-developed MAMbO5 ontology [4]. The MAMbO5 ontology features concepts for modelling IVEs enhanced with concepts for describing agent-based organisational features. The concepts featured in MAMbO5 ontology can be considered abstractions of the concepts that can be found in the GIVEn ontology. The foreseen benefit of this approach is to create a way of modelling the simulated world of a video game, like modelling the real world, thus potentially making it easier to translate simulations

Games in IVE ontology (GIVEn)

intended use

created for the real world into game-based environments. Such an approach is deemed useful since simulated environments are prone to many problems that occur in the real world, such as physical constraints.

The GIVEn ontology is planned to be used for describing video game worlds using the applicable concepts. Such models might be used in simulations or to provide the necessary context to intelligent artificial agents to enhance their reasoning capabilities. Furthermore, a model describing the main concepts of a video game as an IVE can be used to generate various types of content for the modelled game, e.g. to generate worlds based on their ontological models, to generate NPCs that can understand the world, or to automatically generate items that can have features that are aligned with the model of the video game world.

To achieve the described, the GIVEn ontology will be expressed using a high degree of formality, namely classified as rigorously formal by [7]. The formal structure of the ontology, along with its various constraints, guarantees clarity, unambiguity, and ease of use. Additionally, this structure facilitates the process of extending other ontologies or allowing other ontologies to extend it. Lastly, a highly formalized expression of an ontology enhances its compatibility with other digital systems.

Based on the above description of the GIVEn ontology, the ontology is planned to comprise the concepts that specify the concepts that can be found in the MAMbO5 ontology. The specification is geared towards the domain of video games and the concepts necessary to model a video game as an IVE. The foreseen additions will be in terms of specialising the *Agent* concept of the MAMbO5 ontology, possibly the *Artefact* concept, and similar. The GIVEn ontology is not planned to be created to be able to provide concepts for extremely detailed modelling of a video game. Instead, it is planned to provide concepts that can be further specialised and extended towards a specific game genre, video game, or game engine.

The level of granularity stemming from the description above is quite abstract. The ontology should include concepts that can be, for example, used to describe NPCs, monsters and mobs, places, and points of interest in the video game, but another layer of the specification is expected to be added where the concepts of a specific genre of video games should be described and genre-specific entities defined. On top of this domain-specific layer is foreseen the individual-based layer. Therefore, the GIVEn ontology provides generalised concepts that should be further specified.

scenario of use

degree of formality

scope

level of granularity

Knowledge Acquisition

Several ontologies related to video games can be found in publications dating from 2008 or later [8], [9], [10], [11], [12], [13], [14], [15], [16], [17], [18], [19].

One of the earliest frameworks was the Game Ontology Project (GOP) [19]. It was developed as a wiki-based ontology that identifies the important elements of games and their relationships. The outcome was a hierarchical classification of game elements based on the analysis of many specific games. GOP was intended as a tool for game studies, helping scholars converse about games using consistent terminology. At the same time, Digital Game Ontology (DGO) was presented in [9]. This ontology was capable of processing a wide palette of game concepts and in-game events, treating games "as pieces of media" comprising many different elements. The goal of this ontology was to support applications in game production, player activity analysis, and player experience modelling.

Video Game Ontology (VGO) [14] was developed to enable interoperability between games. VGO models the core entities of video games such as players, characters, items, and events, so that information about gameplay can be exchanged across different games. Continuing with the ontologies that are not aimed at being comprehensive in the context of video games, the Ludo ontology [15] was proposed to describe serious games. This ontology provides concepts for describing different aspects of serious games, from educational content to player interactions, and makes it possible for game elements to reference external knowledge bases. Specific ontologies often do not provide concepts for modelling all the aspects of video games. Therefore, the Game Character Ontology (GCO) [17] was introduced to fill one such gap. The GCO is a lightweight ontology for describing video game characters, allowing information about characters from different games or fan communities to be related, integrated, and compared. The authors in [20] present an ontology specifically crafted for role-playing games. This is a good example of a domain ontology in the context of video games. The narrower domain made it possible to define the concepts with greater precision and to capture the nuances of the RPG genre that are not present in other types of video games.

Other than full-fledged ontologies, some taxonomies have been published as well. A classification proposal for serious games was published in [10], featuring 16 criteria for organising and classifying games. This was not the only attempt in standardising game classification. OntoJogo [18] is an ontology-driven classification framework that aims at providing a universal categorisation of games. That framework provides a ‘detailed and consistent definition of a game’ suitable for classifying virtually any game.

The ontology on gameplay, published in [12] provides a vocabulary and model to analyse gameplay situations in terms of player immersion, actions, and how the game responds to them. Providing a universal vocabulary is a goal many game-related ontologies share. One such ontology is the redefined MDA (mechanics, dynamics, and aesthetics) ontology [13]. This ontology is based on and extends the MDA framework, which was originally proposed as a game analysis model [21]. The MDA framework has been criticised for lacking a clear definition of its terms and for not being formalised. The ontology presented in [13] addresses these issues by providing a formal definition of the MDA elements and their relationships.

One of the most recently published ontologies is the one presented in [8], which provides a formal ontology of game spatiality.

The referenced selection of the available publications was consulted, and it was found that only two of the ontologies presented in those publications can be found as publicly available formalised ontologies [14], [16]. Therefore, only the following ontologies are considered moving forward:

- Core Game Ontology (CGO) [16],
- Video Game Ontology (VGO) [14].

These two ontologies are the only ones that can be fully analysed. Completely available ontology files are an important prerequisite to using their concepts in further development since a concept can be fully analysed and understood only when its neighbourhood is available for analysis, too.

Honourable mention is the Game Ontology Project [19], a framework for describing, analysing and studying games. The resulting hierarchy of concepts pertaining to video games is not formalised and is currently available only as a collection of related wiki pages. Another honourable mention is the Game Character Ontology [17], yet the ontology implementation can no longer be found online.

What follows in this section are the definitions and descriptions of the concepts identified as relevant to the context of modelling video games as IVEs that can be found in the chosen ontologies.

These concepts are described based on their definitions in their respective original ontology, although that description, purpose-wise, may change in the further steps of developing the GIVEn ontology. Since the collection of VGO concepts is broader than that of the CGO, only a selection of the VGO concepts is presented here.

⊙ see tables 2.14 – 2.23

⊙ see tables 2.1 – 2.13

In addition to the select concepts of the VGO and CGO ontologies, related directly to the domain of video games, the glossary here contains a selection of concepts from the MAMbO5 ontology that have been identified as possibly relevant to the domain of video games. These concepts are selected as potentially useful when considering video games in the context of IVEs. Their descriptions are cited from [2], [4].

© see tables 2.24 – 2.40

Concept name	Game
Definition	Specifies a video game.
Description	The Game class represents a video game entity within the ontology. It serves as the central hub that connects various facets of a game, such as visuals, audio, narrative, game design, level design, and gameplay. This class encapsulates the essence of a video game, providing a foundation for detailing all its integral components. Through properties like hasVisuals, hasAudio, and hasNarrative, the Game class links to other classes that define specific aspects of the game experience.

Table 2.1: *Game* glossary entry from CGO [16]

Concept name	Visuals
Definition	Visuals contain any visual output of a game, ranging from photorealistic to caricaturized or abstract visuals.
Description	The Visuals class encompasses all the visual elements of a game, including graphics, animations, and visual effects. This class captures the aesthetic essence of the game, whether it portrays realistic landscapes, stylized characters, or abstract art forms. By connecting to the Game class via the hasVisuals property, it specifies the visual characteristics that define the game’s appearance and visual storytelling.

Table 2.2: *Visuals* glossary entry from CGO [16]

Concept name	Audio
Definition	Audio includes background music such as a fully orchestrated soundtrack, sound effects, rewarding sounds, and voice-acted dialogue.
Description	The Audio class covers all auditory components of a game. This includes the musical score that sets the mood, sound effects that provide feedback, and voice acting that brings characters to life. Linked to the Game class through the hasAudio property, it highlights how sound enhances the gaming experience by adding depth and immersion.

Table 2.3: *Audio* glossary entry from CGO [16]

Concept name	Narrative
Definition	Narrative contains the interactive story of a game, which makes up the game's plot.
Description	The Narrative class represents the storyline and plot elements within a game. It includes the characters, events, and dialogues that engage the player in the game's universe. Through the hasNarrative property, it connects to the Game class, emphasizing the importance of storytelling in creating an immersive and emotionally compelling experience.

Table 2.4: *Narrative* glossary entry from CGO [16]

Concept name	GameDesign
Definition	Game design contains all the game's mechanics that define the game's rules, providing structures and frames for play (e.g., winning and losing conditions) and actions available to the player.
Description	The GameDesign class focuses on the underlying mechanics and rules that govern gameplay. It outlines how players interact with the game world, the challenges they face, and the goals they strive to achieve. By linking to the Game class via the hasGameDesign property, it details the structural framework that shapes the player's experience and defines what makes the game engaging and challenging.

Table 2.5: *GameDesign* glossary entry from CGO [16]

Concept name	LevelDesign
Definition	Level design includes the architecture of the spatial navigation of levels, determining how the player can progress from one point in the game to another.
Description	The LevelDesign class pertains to the creation of game environments and levels. It involves the layout of spaces, placement of obstacles, and pathways that guide player progression. Connected to the Game class through the hasLevelDesign property, it highlights how thoughtful level design enhances exploration, pacing, and the overall enjoyment of the game.

Table 2.6: *LevelDesign* glossary entry from CGO [16]

Concept name	Gameplay
Definition	Gameplay consists of the player’s strategies while playing a game.
Description	The Gameplay class represents the interactive experience from the player’s perspective. It encompasses the strategies, decision-making processes, and play styles that players adopt. Through the hasGameplay property, it links to the Game class, illustrating how player interaction with game mechanics leads to a unique and personal experience for each individual.

Table 2.7: *Gameplay* glossary entry from CGO [16]

Concept name	hasVisuals
Definition	Specifies the visuals of a game.
Description	The hasVisuals property connects a Game to its Visuals, indicating the visual content that defines the game’s aesthetic. This relationship showcases how visuals contribute to the game’s identity and player immersion, whether through stunning graphics or distinctive art styles.

Table 2.8: *hasVisuals* glossary entry from CGO [16]

Concept name	hasAudio
Definition	Specifies the music and sound of a game.
Description	The hasAudio property links a Game to its Audio, detailing the auditory elements that enhance the gameplay experience. This includes everything from ambient sounds and musical scores to character voices, all of which contribute to the game’s atmosphere and emotional impact.

Table 2.9: *hasAudio* glossary entry from CGO [16]

Concept name	hasNarrative
Definition	Specifies the story and plot of a game.
Description	The hasNarrative property associates a Game with its Narrative, outlining the storylines, character arcs, and plot developments that engage players. This connection underscores the role of storytelling in creating meaningful and memorable gaming experiences.

Table 2.10: *hasNarrative* glossary entry from CGO [16]

Concept name	hasGameDesign
Definition	Specifies the rules and mechanics of a game.
Description	The hasGameDesign property links a Game to its GameDesign, highlighting the mechanics that define how the game is played. This includes rules, controls, objectives, and other elements that shape the player’s interaction with the game world.

Table 2.11: *hasGameDesign* glossary entry from CGO [16]

Concept name	hasLevelDesign
Definition	Specifies the levels of a game.
Description	The hasLevelDesign property connects a Game to its LevelDesign, focusing on the spatial and environmental aspects of the game. This relationship emphasizes how level design affects gameplay dynamics, exploration, and the player’s journey through the game.

Table 2.12: *hasLevelDesign* glossary entry from CGO [16]

Concept name	hasGameplay
Definition	Specifies the gameplay of a game.
Description	The hasGameplay property links a Game to its Gameplay, emphasizing the player’s experience and interaction with the game mechanics. It reflects how the game engages players, challenges them, and allows them to develop strategies and skills.

Table 2.13: *hasGameplay* glossary entry from CGO [16]

Concept name	Achievement
Definition	An achievement is a reward gained in a game due to some event accomplished in the game.
Description	The Achievement class represents rewards that players earn by completing specific tasks, reaching milestones, or exhibiting particular skills within a game. These achievements serve as incentives and recognition for players, enhancing engagement and providing goals to strive for. Achievements are linked to Games through the hasAchievement property, indicating which rewards are available in each game. Players can own achievements ownsAchievement, and they can be earned during gameplay sessions involvesAchievement, enriching the player’s gaming experience.

Table 2.14: Achievement glossary entry from VGO [14]

Concept name	Character
Definition	Any actor that can exist in a game.
Description	The Character class encompasses all entities that act within the game world, including player-controlled avatars and non-player characters NPCs. Characters can be humans, animals, vehicles, or even abstract representations like paddles in Pong. They are central to gameplay, influencing how players interact with the game. Characters are associated with Games via the hasCharacter property, and players can own characters ownsCharacter. During gameplay sessions, characters participate and may trigger events isEventTriggeredByCharacter, highlighting their dynamic role in the game.

Table 2.15: Character glossary entry from VGO [14]

Concept name	Game
Definition	Describes a game product that can be played by a player.
Description	The Game class represents the game itself as a product and interactive experience. It serves as a hub linking to various components such as Achievements, Characters, Items, Genres, PlayingAreas, and Leaderboards. Through properties like hasAchievement and hasCharacter, it details the content and features that define the game. Players interact with games playsGame, and sessions Session represent periods during which the game is played, capturing the dynamic nature of gameplay.

Table 2.16: Game glossary entry from VGO [14]

Concept name	Genre
Definition	Describes the genre a game belongs to.
Description	The Genre class categorizes games based on their style, mechanics, and thematic elements, such as RPGs, simulators, or adventure games. By linking Games to their genres through the hasGameGenre property, it provides context and helps players understand what to expect from the game, aiding in discovery and classification.

Table 2.17: *Genre* glossary entry from VGO [14]

Concept name	Item
Definition	Portrays any item that exists in a game.
Description	The Item class includes all objects that can be found or used within a game, ranging from weapons and tools to collectibles and decorative objects. Items are integral to gameplay, often enhancing a character’s abilities or serving as objectives. They are connected to Games via the hasItem property and can have specific features hasFeature. Characters can own items ownsItem, and items may be involved in events isEventRelatedToItem, illustrating their multifaceted role in the game.

Table 2.18: *Item* glossary entry from VGO [14]

Concept name	Leaderboard
Definition	Describes a ranking system of the players.
Description	The Leaderboard class represents competitive ranking systems that track player performance, such as high scores, fastest times, or most victories. Leaderboards encourage competition and replayability, fostering a sense of community. They are linked to Games through the hasLeaderboard property, indicating where players can compete and compare achievements.

Table 2.19: *Leaderboard* glossary entry from VGO [14]

Concept name	Player
Definition	The entity playing the game.
Description	The Player class represents individuals who engage with the game, whether human players or AI-controlled agents. Players are central to the gaming experience, bringing the game to life through interaction. They have profiles that may include usernames <code>username</code> , real-world locations <code>livesIn</code> , and relationships with other players <code>isFriendWithPlayer</code> . Players own characters <code>ownsCharacter</code> and achievements <code>ownsAchievement</code> , participate in sessions <code>involvesPlayer</code> , and their actions can trigger events <code>isEventAssociatedToPlayer</code> , capturing the richness of the player experience.

Table 2.20: *Player* glossary entry from VGO [14]

Concept name	Session
Definition	Describes a session of gameplay.
Description	The Session class captures specific instances of gameplay, such as a match, level, or any defined period during which the game is played. Sessions are linked to Games <code>isSessionInGame</code> , involve Players <code>involvesPlayer</code> and Characters <code>involvesCharacter</code> , and can include events and achievements earned during that time <code>involvesAchievement</code> . Sessions provide a framework for analyzing gameplay, understanding player behavior, and tracking progress over time.

Table 2.21: *Session* glossary entry from VGO [14]

Concept name	PlayingArea
Definition	Describes the place where the gameplay takes place.
Description	The PlayingArea class defines the environments or settings within a game where the action unfolds. This could be a specific level, map, world, or any spatial context like a racetrack, battlefield, or fantasy realm. Through the <code>hasPlayingArea</code> property, Games are connected to their playing areas, emphasizing the importance of environment in shaping gameplay and player immersion.

Table 2.22: *PlayingArea* glossary entry from VGO [14]

Concept name	InstantaneousEvent
Definition	An event that happens during gameplay at a certain moment in time.
Description	The InstantaneousEvent class encompasses events that occur at specific moments during gameplay, triggered by player actions, game mechanics, or other factors. These events can include earning an achievement <code>unlocksAchievement</code> , defeating an enemy, or making a purchase <code>InAppPurchaseEvent</code> . Instantaneous events are linked to Games <code>isEventInGame</code> , Sessions <code>isEventInSession</code> , Players <code>isEventAssociatedToPlayer</code> , Characters <code>isEventTriggeredByCharacter</code> , and Items <code>isEventRelatedToItem</code> , providing a detailed account of the dynamic occurrences within the game.

Table 2.23: *InstantaneousEvent* glossary entry from VGO [14]

Concept name	Action
Definition	An action is the building block of agents' activities.
Description	An action is essentially an agent's response to tasks. Whereby tasks are created to be met or reached, an action is the atomic concept for achieving tasks. In the context of this document, an action is the building block of a process, and agents' ability to act towards its environment in general. Every action can be used to fulfill at least one task.

Table 2.24: *Action* glossary entry from MAMbO5 [4]

Concept name	Agent
Definition	A piece of software that can act upon its environment and perceive it.
Description	An agent in the context of this document is a piece of software that can interact with its environment, act upon it, and, in case of an intelligent agent, reason upon their accessible knowledge. Indeed, an agent is <i>anything that can be viewed as perceiving its environment through sensors and acting upon that environment through actuators</i> . [22] In the organisational context of this document, a software agent is essentially a model of a real-life person.

Table 2.25: *Agent* glossary entry from MAMbO5 [4]

Concept name	Artefact
Definition	An artefact is an otherwise unclassified element of an organisation system.
Description	An artefact is, as of yet, a somewhat undefined concept, in the context of specifying its domain. Essentially, an artefact can be anything that is not classified using the other classes of this ontology. Furthermore, an artefact can be physically representative (e.g. a chair), or an unphysical concept (e.g. knowledge). Artefacts therefore represent various concepts that the agents can interact with, or that affect the given environment or the given system, i.e. objects forming the environment.

Table 2.26: *Artefact* glossary entry from MAMbO5 [4]

Concept name	Behaviour
Definition	A behaviour is an implemented action, or a part of one, that defines how an agent interacts with its environment or internal states.
Description	In the context of this document, a behaviour is a set of actions or operations that an agent can perform in response to its environment or internal states. Behaviours are essential components that define how an agent acts and reacts, allowing it to achieve its objectives and respond to changes. Moreover, a behaviour is, in this context, a way of implementing a specific action and is that which an agent can actually run in runtime.

Table 2.27: *Behaviour* glossary entry from MAMbO5 [4]

Concept name	Role
Definition	A role is a set of norms with a common denominator.
Description	In the context of this document, a role is defined as a set of normative rules that are applicable to a particular part of the given organisation. Such normative rules are parts of the organisation's normative system, and can be grouped by specific criteria, thus forming roles. Roles are played by agents. When an agent plays a role, the role's constraints are applied to them, therefore constraining their possible actions, their perceivable goals, and their possibilities in general.

Table 2.28: *Role* glossary entry from MAMbO5 [4]

Concept name	Workspace
Definition	A workspace is the union of all the elements of a system, including agents, artefacts, etc.
Description	A workspace is the complete environment of a given system, including all the agents, artefacts, etc. What sets the concept of a workspace apart from the concept of an environment is the extent of the involved concepts, i.e. a workspace contains all the elements of an organisation and the whole system, while environment comprises only the elements that are external to the given organisation. It is worth noting that elements of the environment are an integral part of the whole system, since the life and activities of the given organisation are influenced by them.

Table 2.29: *Workspace* glossary entry from MAMbO5 [4]

Concept name	Intelligent Virtual Environment
Definition	An intelligent virtual environment is a virtual environment that simulates the real world, and is populated by autonomous intelligent entities. [5]
Description	Intelligent virtual environments are researched as an area on the intersection of two aspects pertaining to the concept of artificial intelligence, if only but marginally: intelligent tools and techniques that are embodied in autonomous agents (real-life and digital alike), and effective ways of representing them, along with various means of achieving different kinds of interaction amongst them [5], [6]. In other words, a IVE is a concept that represents a virtual environment whose main goal is simulating a segment of the real world, populated by artificial autonomous entities (agents). [5]

Table 2.30: *Intelligent Virtual Environment* glossary entry from MAMbO5 [4]

Concept name	IVE Law
Definition	A IVE law is a norm that is valid only within a specified physical space (a IVE workspace).
Description	A special kind of a norm, an IVE law is a norm that is constrained by its applicability to a specific physical space, i.e. a specific IVE workspace. Being applicable to only a restricted area means that every IVE law is valid only within the bounds of the given area (a IVE workspace), and never outside of that specified space. This kind of a norm is the key constraint of the concept of a situated organisational unit.

Table 2.31: *IVE Law* glossary entry from MAMbO5 [4]

Concept name	IVE Workspace
Definition	
Description	Complimentary to the concept of a workspace, a IVE workspace represents a physical location, or a physically describable location.

Table 2.32: *IVE Workspace* glossary entry from MAMbO5 [4]

Concept name	Objective
Definition	An objective is a high-level goal the be met, suitable for the context of strategic planning.
Description	An objective is more general than a goal, although their definitions are rather similar. Fulfilling several goals can lead an organisational unit towards fulfilling a set objective. Thus, an objective is more suitable in the context of strategic planning, while a goal is more suitably used in the context of short-term planning.

Table 2.33: *Objective* glossary entry from MAMbO5 [4]

Concept name	Observable Property
Definition	An observable property is a property of an artefact that can be observed by agents in the same IVE.
Description	This is a property of an artefact located in a IVE that is observable by other agents located within the same IVE. These are tightly connected to the concept of observable events, and can be influenced upon by an operation.

Table 2.34: *Observable Property* glossary entry from MAMbO5 [4]

Concept name	Physical Artefact
Definition	Physical artefacts are all the concepts that can be physically represented and included in a IVE.
Description	Every concept that describes objects that can be physically represented (e.g. a top hat), i.e. embodied and positioned on a topological map, and as such included in a IVE are classified as physical artefacts. Such elements have their role to play in the given IVE and usually contain a defined interface that governs the process of interaction of an agent with the given physical artefact.

Table 2.35: *Physical Artefact* glossary entry from MAMbO5 [4]

Concept name	Physical Property
Definition	
Description	Physical properties are key elements of physical artefacts, i.e. artefacts that can be visualised in a physical space. Usually when an artefact is used, a physical event is generated, and a physical property is modified.

Table 2.36: *Physical Property* glossary entry from MAMbO5 [4]

Concept name	Goal
Definition	A goal is a result towards which effort is directed - an end to be met.
Description	A goal is broadly defined as a result or achievement towards which effort is directed ¹ . In the context of this document, a goal is a form of an objective. A goal is an end to be met or reached, and can consist of several sub-goals.

Table 2.37: *Goal* glossary entry from MAMbO5 [4]

Concept name	Quest
Definition	A quest is similar to a goal, but has a defined starting and ending situations.
Description	A quest is a similar to a goal, but it has a defined beginning and a defined end, i.e. a starting situation, and an ending situation ² . In the context of cro:MMORPGmassively multi-player on-line cro:RPGrole-playing games (MMORPGs), a quest is what drives a story, and, in principle, motivates the player to continue playing the game. Furthermore, a quest is often given to the player by an in-game character. A quest usually has various stages, and represents a challenge for the given player, thus embarking them on an adventure.

Table 2.38: *Quest* glossary entry from MAMbO5 [4]

Concept name	Strategic Alliance
Definition	Strategic alliance is a form of a long-lasting partnership of organisations of various forms, formed around a shared strategy, or a strategic goal.
Description	An alliance that is aimed at forming long-lasting partnerships consisting of organisations of various forms is dubbed a strategic alliance. A strategic alliance is formed around a strategy as a long-term objective that is shared amongst the strategic alliance members. Norms and regulations governing the expected behaviour within the strategic alliance are expected to be accepted by all the members, old and new alike.

Table 2.39: *Strategic Alliance* glossary entry from MAMbO5 [4]

Concept name	Task
Definition	A task is the building block of a quest.
Description	A task is the building block of a quest, i.e. its elementary part. A quest is built of atomic tasks that are easier to follow in execution phase, rather than the overview provided by the main definition of a quest. In MMORPGs a quest could demand an item to be retrieved, yet such a simple-sounding quest could consist of various tasks that have to be fulfilled in order for the main quest to be finished. The relation of quest and task concepts can be recursive ³ .

Table 2.40: *Task* glossary entry from MAMbO5 [4]

3

Conceptualisation

The acquired knowledge laid out in the previous section is transformed into a structured and meaningful model in this phase of ontology development. During this phase, therefore, the key concepts, relationships, attributes, and rules of the domain are identified and organized to reflect the domain's structure. The goal is to create a clear and coherent conceptual model that captures the domain's essential elements without formalizing them into specific ontology languages. The conceptual model produced in this phase is not yet formalized and can be rather unstructured, but provides a foundational blueprint for the subsequent formalization and implementation phases of the chosen ontology development process.

First, a graphical representation of the concepts and their relationships is created. The concepts are represented as nodes, and the relationships between them as edges. Not all of the relationships are to be modelled as properties in the ontology afterwards. Before proceeding to the formalisation phase, the graphical representation is reviewed and refined to ensure that it accurately captures the domain's structure and that all relevant concepts and relationships are included. Furthermore, it is possible that some of the concepts are not necessary after all, and can be discarded when moving on to the formalisation phase. On the other hand, additional concepts may be identified, that have not been a part of the acquired knowledge presented in the previous section, nor in the list of concept provided in it, but should be added to the graphical representation.

© see figure 3.1

The prepared visual representation of the conceptual model is based on the concepts of three ontologies. The concepts from the MAMbO5 ontology are represented in green, those from the CGO in blue, and those from the VGO in orange. The concepts that are additions to the ones found in the ontologies are represented in dashed boxes. The concepts that are not going to be included in the final ontology are absent from the visual conceptual model.

Conceptual modelling was started using the fundamental concepts of MAMbO5 ontology, such as Agent, Role, Behaviour, Action, Strategy, Artefact, Knowledge Artefact, Norm, and Workspace. The concepts from the CGO and VGO were added later, as they were deemed relevant to the domain of video games and their representation as an IVE. Those concepts from CGO and VGO that

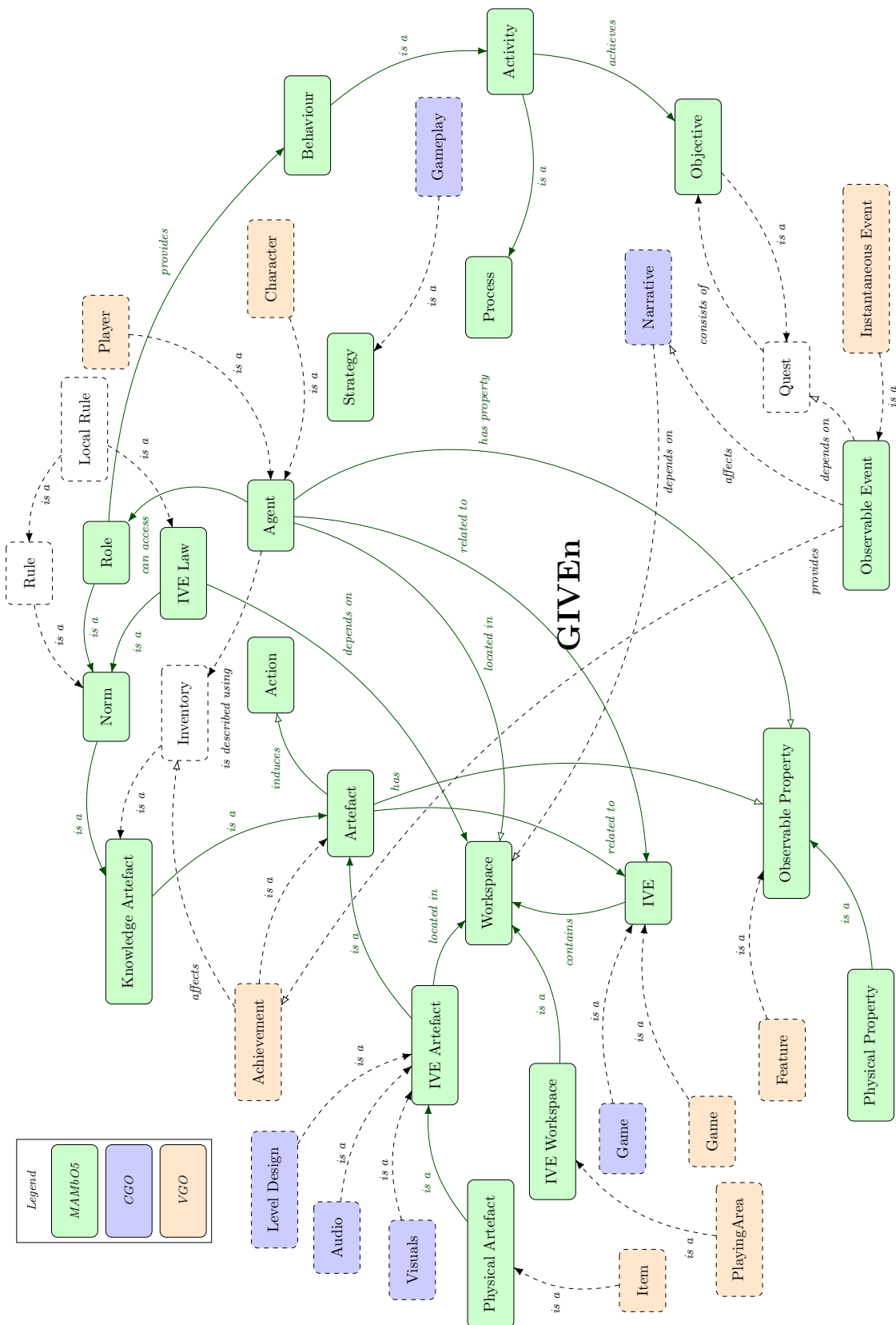


Figure 3.1: Concepts and their relationships in the GIVEN conceptual model

were left in the model are represented in blue and orange, respectively, in dashed boxes, indicating that they are additions to the ones found in the MAMbO5 ontology. The additional concepts, not previously found in any of the three observed ontologies, were added as the need arose, and are represented in clear dashed boxes in the visual representation of the conceptual model.

The **Agent** concept is one of the central concepts in the model, as it represents the entities that are bound to a specific video game and can interact with the in-game world, i.e. some **Workspace**, and with each other. The **Agent** can be either a **Player** or a **Character**. The **Player** represents the human user in a video game, while the **Character** represents the NPCs in the game. The **Character** concept can be further divided into subconcepts representing specific types of NPCs, yet that division is not to be present in the current version of the ontology, but is expected to be expanded with in the future, i.e. should be added when using the subject on ontology for modelling a specific video game genre.

Each **Agent** can access a set of **Role** individuals that are relevant within the context of the video game they are related to, and wherein they can be found. The **Role** concept represents the different roles that the agents can play in the game, and is a key element in the game's narrative and gameplay. Roles are especially popular in *cro:RPG*-playing games (RPGs), where the player can choose from a set of predefined roles, each with its own set of abilities and limitations. The **Role** is associated with a set of **Behaviour** individuals that define the actions and activities that an agent can perform when playing that role. In other words, behaviours are the means by which the agents can achieve the objectives set by the roles they play, i.e. behaviours are a way that agents can affect the environment they are in. The **Behaviour** are associated with **Action** that are the actual actions that the agent can perform. Several actions may be grouped together to form a **Behaviour**, and a single action may be a part of multiple behaviours. For example, a behaviour of shooting a bow may be composed of the actions of drawing the bow, aiming, and releasing the arrow. On the other hand, the action of drawing the bow may be a part of the behaviour of shooting a bow, but also of the behaviour of reloading the bow, or the behaviour of aiming the bow.

Since **Action** class is defined as atomic by MAMbO5, it is not further divided into subconcepts in the current version of the ontology. However, it is expected that the ontology will be expanded with in the future, i.e. more specific concepts may be added when using the ontology for modelling a specific video game genre. Activities can be used to achieve certain **Objective** class individuals. These objectives can be observed as specific subclasses of the **Quest** class. An objective should be described here as an individual goal to be accomplished, while a quest is a collection of objectives that are to be accomplished in a specific order. A quest may be specific to some particular game genres, but it may be considered in a more

general and abstract manner as a list of objectives with a specific starting situation and a specific ending situation. The quest is a way to structure the gameplay and provide the player with a sense of progression and accomplishment. The quest is also a way to structure the narrative of the game, as the player's progress through the quest can be used to drive the story forward. The quest is, therefore, a way to connect the gameplay and the narrative of the game.

The concept of `Gameplay` represents the way a player interacts with a given game. Therefore, this concept is a part of the `Strategy` concept, as the player's strategies are the means by which they interact with the game and achieve their objectives. The `Strategy` concept is a way to represent the player's approach to the game, and can be used to model different play styles and strategies that players can use to achieve their objectives.

The concepts of a `Rule` and `LocalRule` are added to the model to represent the game's mechanics. The `Rule` concept represents the general rules of the game, while the `LocalRule` concept represents the rules that are specific to a particular quest or a particular part of the game. The `Rule` and `LocalRule` concepts are both subclasses of the `Norm` concept, since they are both normative rules that define the behaviour of the agents in the game. The `Rule` is a more general concept, as it represents the rules that are applicable in the entire game, while the `LocalRule` is a more specific concept that represents the rules that are applicable only in a specific context, i.e. a specific location in the IVE. The `Rule` concept is a subclass of `Norm`, just like the concept of `Role`, but while the `Role` concept represents the roles that the agents can play in the game, the `Rule` concept represents the rules that govern more than just the behaviour of the agents in the game. While roles empower agents, rules constrain the world they are in.

A collection of norms (including roles and rules) is a specific type of `Knowledge Artefact` concept, since they are a piece of knowledge that is shared among agents within an organisation. Furthermore, such a collection may be queried by the agents to determine the applicable norms in a specific context. This concept is a subclass of the `Artefact` concept, and represents a piece of knowledge or information that is specific to an individual agent or shared among agents within an organisation. More specifically, a knowledge artefact may contain knowledge that can be retrieved by individual agents. A specific new concept was added to the conceptual model to represent an artefact containing a collection of data about a single agent, i.e. the `Inventory` concept. The `Inventory` concept represents the data that is specific to a single agent, and is therefore a specific type of a `Knowledge Artefact`. Although the concept of inventory is used by video games to store specific in-game items that a player or a character can carry with them, the `Inventory` concept is a more general concept that can be used to store any data that is specific to a single agent, such as their health, level, score, or any other relevant information. There is no restriction on how the

data is stored in the `Inventory` concept, and it is up to the specific genre implementation of the ontology to decide how the data is structured and represented.

One specific form of an artefact defined in this conceptual model is the VGO concept `Achievement`. The `Achievement` represents a reward that is given to a player for accomplishing a certain task in the game. Finishing a task should be modelled as an `Observable Event`, as it is an event that can be observed by the agents in the game. An achievement may affect several different elements of a game, yet the conceptual model shows its most common effect, i.e. that it affects an agent's `Inventory`. For example, an achievement might be a badge the player will receive, a specific number of experience points, a power-up that can be used in the game, or an `Item` awarded to the player participating in an `Observable Event`.

Concepts `Visuals`, `Audio`, and `LevelDesign` are added to the model to represent the different aspects of the game's aesthetics, i.e. various assets related to a game. The `Visuals` concept represents the visual elements of the game, such as the graphics and the user interface. The `Audio` concept represents the audio elements of the game, such as the music and the sound effects. The `LevelDesign` concept represents the design of the levels in the game, i.e. the layout of the game world and the placement of the in-game elements. Most of the objects individuals of these concepts are located in the `Workspace` concept, as they are the elements that make up the game world and are related to specific areas or maps or other instances of the `PlayingArea` concept.

One of the concepts that should not be excluded is the concept of an `Item`. Since items are usually rendered as physical, relative to the in-game world, they are modelled as a subclass of the `Physical Artefact` concept. It should be noted here that artefacts have related `Action` individuals, as they are the means by which the agents can interact with the artefacts. For example, an `Item` can be picked up by an agent, or used in some way. The `Action` individuals related to an artefact represent the ways in which the agents can interact with the artefact.

As opposed to an `Inventory` individual describing and defining some aspects of an agent, `Observable Property` individuals are used to define the properties of the artefacts that may be observed by agents. For example, an `Item` may have a property that defines its weight, or its value, or its rarity. These properties can be observed by agents, and can be used to determine the behaviour of the agents towards the artefact. Specific properties that define what an artefact looks like or where it is located are modelled as `Physical Property` individuals, as they are properties that are specific to the physical representation of the artefact in the game world.

According to some of the most popular literature on video

games, there are four basic elements that form a game [23]:

1. **Mechanics** – create the structure of the game and define how players interact with it;
2. **Story** – provides context and meaning to the gameplay experience;
3. **Aesthetics** – contributes to the game’s atmosphere and style;
4. **Technology** – used to create and run the game.

These elements are present in the proposed conceptual model as well, even though several concepts of the model may be mapped to a single element. For example, the `Rule` and `LocalRule` concepts, as well as the `Role` concept, can be considered as part of the mechanics of the game. The `Visuals`, `Audio`, and `LevelDesign` concepts can be considered as part of the aesthetics of the game. The `Narrative` concept can be considered as part of the story of the game. The technology element is not explicitly represented in the conceptual model, as it is not relevant to the domain of video games and their representation as an IVE, i.e. the technology element includes the game engine, the programming language, the graphics engine, the audio engine, etc., which are not deemed relevant to the domain of the ontology.

© see figure 3.2

Concept name	Inventory
Definition	Inventory is the dynamic collection of items, resources or tokens currently owned or controlled by an agent
Description	The inventory acts as a virtual container that stores all the data relevant to an agent, including items, resources, tokens, and other data.

Table 3.1: *Inventory* concept details

Concept name	Quest
Definition	A quest is a structured mission composed of one or more objectives that a participant must accomplish to progress and earn rewards within the system
Description	In gamified multi-agent environments quests translate higher-level goals into actionable steps that guide motivational flow through clear challenges feedback and reward strategies. They can be dynamically generated to adapt to the user profile or system state and often reference narrative elements inventory updates and achievement provisioning to maintain engagement and learning momentum

Table 3.2: *Quest* concept details

Concept name	Rule
Definition	A rule is a formal statement that specifies how entities may act or how the system responds to particular conditions
Description	Rules encode the logic that drives autonomous agent decisions validation of user actions and state transitions. They can reference observable properties define penalties or rewards and trigger narrative or inventory changes. Rules may be global, applying across the whole system, or local when scoped to a workspace ensuring predictable yet flexible behaviour

Table 3.3: *Rule* concept details

Concept name	Local rule
Definition	A local rule is a constraint or operational instruction that applies only within a specific workspace or context of the multi-agent gamified system
Description	Local rules tailor system behaviour to the conditions of a particular environment or scenario. They can override or extend global rules enabling fine-grained control over interactions resources and feedback loops. In gamification they allow designers to create context-sensitive mechanics such as area-based bonuses or restrictions that foster adaptive and immersive experiences

Table 3.4: *Local rule* concept details

Concept name	is described using
Definition	This property associates an entity with the descriptive artefact that is inventory
Description	This property joins an agent to its inventory. The inventory contains all the data relevant to the agent, including items, resources, tokens, and other data. Therefore, the inventory is the descriptive artefact that describes the agent.
Domain	Agent
Range	Inventory

Table 3.5: *is described using* concept details

Concept name	affects inventory
Definition	Affects inventory connects a rule quest or event to the inventory whose contents it alters
Description	By declaring this relationship designers can track and reason about the flow of resources and items ensuring that system audits and balance checks consider every mechanic that adds removes or transforms inventory elements. It supports analytics on economic impact and dependency mapping.
Domain	Achievement
Range	Inventory

Table 3.6: *affects inventory* concept details

Concept name	affects narrative
Definition	This property links an entity of class Observable event to the narrative element whose storyline or exposition it modifies
Description	This property enables the system to maintain narrative coherence by explicitly declaring the causal influence of mechanics on storytelling fragments. When a rule fires or a quest is completed the narrative arc referenced here can branch unlock or update which maintains immersion and personalised plot evolution.
Domain	Observable Event
Range	Narrative

Table 3.7: *affects narrative* concept details

Concept name	consists of objectives
Definition	Consists of objectives connects a quest to the discrete objectives that compose it
Description	The property decomposes complex missions into measurable sub-goals enabling adaptive pacing progress tracking and personalised support. It supports hierarchy creation so that tools can visualise nested objectives and calculate completion percentages.
Domain	Quest
Range	Objective

Table 3.8: *consists of objectives* concept details

Concept name	depends on
Definition	Depends on expresses a prerequisite relationship where the source entity requires the completion existence or truth of the target entity
Description	Typical applications include modelling task precedence quest chains or conditional rule activation that foster logical progression and prevent conflicts. The property supports transitive reasoning which allows agents to infer readiness and plan accordingly.
Domain	Observable Event
Range	Quest

Table 3.9: *depends on* concept details

Concept name	depends on workspace
Definition	Depends on workspace relates an entity to the workspace context that must be active for the entity to be valid or executable
Description	Workspaces represent distinct functional or spatial zones such as a classroom module or level. By binding quests and rules to a workspace the ontology captures situational relevance and prevents unintended cross-context interactions which promotes modular design and easier deployment in distributed settings.
Domain	Narrative
Range	Workspace

Table 3.10: *depends on workspace* concept details

Concept name	has observable property
Definition	Has observable property links an entity to a property that can be sensed or measured in the environment
Description	The relation allows rules and agents to monitor state variables such as time score or physiological signals which facilitates context-aware adaptations and evidence-based feedback. It supports integration with Internet of Things sensors analytics dashboards and self-regulated learning loops.
Domain	Agent
Range	Observable Property

Table 3.11: *has observable property* concept details

4

Formalisation

The formalisation phase in ontology engineering focuses on applying formal semantics to the conceptual model of a domain. This phase aims to create an ontology that is precise, unambiguous, and capable of supporting complex reasoning. By using a formal language, the ontology's structure, relationships, and constraints become systematically interpretable by both humans and machines.

The formalisation of the GIVEn ontology is based on the Web Ontology Language (OWL). OWL is a family of knowledge representation languages for authoring ontologies. OWL ontologies are based on the Resource Description Framework (RDF) and are designed to be machine-readable and machine-interpretable. Even so, GIVEn ontology is given a human-readable documentation in this deliverable, using Turtle syntax serialization, as well as a machine-readable formalisation in the OWL format.

© see GIVEn Turtle Serialization

5

Integration

The integration phase of this ontology engineering methodology focuses on harmonizing the developed ontology with other relevant ontologies or knowledge sources to create a unified knowledge ecosystem. This phase enriches the ontology's content, enhancing its semantic reach and improving its reusability across various applications.

Connecting with external ontologies supports data consistency, reduces redundancy, and ensures that the developed ontology can communicate effectively within broader systems, ranging from semantic web applications to intelligent multi-agent frameworks.

The GIVEn ontology is designed as an extension of the following ontologies [4], [16], [24]:

- MAMbO5 – an ontology for multiagent systems, which provides the basic concepts for describing agents, their roles, and their behaviour;
- CGO – a video game ontology, which provides the concepts for describing video games, their elements, and their dynamics;
- VGO – a video game ontology, which provides the concepts for describing video games, their elements, and their dynamics.

The GIVEn ontology is designed to be general enough to be applicable to any video game, yet specific enough to capture the essential elements of video games as IVEs. Therefore, the ontology is designed as an extension of the CGO and VGO ontologies, which provide the concepts for describing video games, their elements, and their dynamics. The MAMbO5 ontology is used as a basis for describing the agents and their behaviour in the video game, as well as the roles they play.

One of the goals of having developed this ontology is to provide a common ground for describing video games as IVEs that can be used in various applications. Therefore, the GIVEn ontology is designed to be compatible with other ontologies and knowledge sources, such as the ontologies for describing gamification and gamified systems. The compatibility is achieved by reusing the concepts from those ontologies and adapting them to the context of video games as IVEs.

6

Implementation and Evaluation

6.1 Implementation

The implementation phase of the ontology engineering process involves creating a concrete representation of the ontology in a specific format or tool. This phase ensures that the ontology is not only theoretically sound but also practically usable. The implementation phase may include tasks such as developing software tools for managing and querying the ontology, creating user interfaces for accessing the ontology's knowledge, and integrating the ontology with existing systems or applications.

The GIVEn ontology is implemented using the Protégé tool for engineering ontologies. The implemented ontology is available on the GitHub repository of this project and is OWL/XML serialized in appendix.

The implemented ontology is consistent and provides additional knowledge when reasoned upon using one of the reasoners available in Protégé or `owlready2`.

© see GIVEn OWL/XML Serializa-
tion

6.2 Evaluation

The GIVEn ontology was evaluated by the domain experts from the host institution using the applicable interview and review approaches. Furthermore, the ontology was tested against the specification document.

© see chapter 1

Part II

The Gamification of an Intelligent Virtual Environment Ontology MAGO-Ga

1

Specification

This chapter presents the ontology engineering process for the MAGO-Ga ontology. The process is based on the METHONTOLOGY methodology [1]. The METHONTOLOGY methodology is a structured approach to ontology development that consists of six main steps: specification, knowledge acquisition, conceptualisation, formalisation, integration, and implementation.

The specification phase is the first step in the ontology engineering process. The goal of this phase is to define the purpose, scope, and intended use of the ontology. The specification phase also involves identifying the key concepts and relationships that will be included in the ontology.

The MAGO-Ga ontology is designed to model gamified multi-agent systems modelled as intelligent virtual environments (IVEs). The main objective of the ontology is to provide a formal representation of the gamification elements and their interactions with the agents in the system. In addition to the gamification elements, the ontology is also aimed at representing personality traits and facets based on the Five Factor personality model [25]. Finally, this ontology is aimed to serve as a knowledge base for reasoning about gamified multi-agent systems.

The intended use of the ontology is to represent the gamification elements and their interactions with the agents in the system. The ontology will also be used to represent the agents' personalities and their interactions with the gamification elements. In order to make it possible to model how the gamification elements affect the agents' personalities, the ontology will include concepts related to the Five Factor personality model. It should be noted here that this ontology does not aim to comprise all the concepts related to gamification, but rather a subset of concepts that are applicable to gamified multi-agent systems. Therefore, only a specialised selection of concepts of the related ontologies will be considered for inclusion in the MAGO-Ga ontology.

intended use

The modelled ontology can be used for modelling a specific gamified multi-agent system, but it is not intended to model individual agents or their specific interactions. The ontology is rather intended to model the system's design, i.e. the gamification elements and their interactions with the agents in the system, and

the agents' personalities and the personalities specific gamification techniques influence the most.

A proposed scenario of use of the MAGO-Ga ontology is to model a gamified multi-agent system for social simulations. Agents are expected to have personalities that are represented using the Five Factor personality model. The gamification elements are expected to influence the agents' behaviour or decisions, and the agents' personalities are expected to influence their interactions with the gamification elements.

scenario of use

The MAGO-Ga ontology will be expressed using a high degree of formality, namely classified as rigorously formal by [7]. Such a formal structure and its many constraints ensure that the ontology is clear, unambiguous, and easy to use. Furthermore, it is easier to use it to extend other ontologies or be extended by other ontologies. Lastly, a highly formalized expression of an ontology makes it easier to use it in conjunction with other digital systems.

degree of formality

The intended level of granularity is expected to make it possible to model a specific gamified multi-agent system, but not to model individual agents or their specific interactions. Furthermore, the ontology is not intended to model the implementation of the system, but rather the system's design.

level of granularity

Based on the above description of the MAGO-Ga ontology, the ontology is planned to comprise the concepts related to gamification, personality traits and facets, and the interactions between them. The ontology will be based on the existing ontologies, but only a specialised selection of concepts will be considered for inclusion in the MAGO-Ga ontology.

scope

Knowledge Acquisition

The main focus of the knowledge acquisition process involved in the making of this ontology is the acquisition of knowledge related to gamification. The knowledge acquisition process is based on the METHONTOLOGY methodology [1]. The METHONTOLOGY methodology is a structured approach to ontology development that consists of six main steps: specification, knowledge acquisition, conceptualisation, formalisation, integration, and implementation.

The knowledge acquisition process is the second step in the ontology engineering process, although it is an ongoing process. The goal of this phase is to identify the key concepts and relationships that will be included in the ontology. The knowledge acquisition process involves researching the existing ontologies applicable to describing gamification, and identifying the most relevant concepts and relationships. The knowledge acquisition process also involves identifying the most relevant sources of knowledge, such as expert interviews, text analysis, and brainstorming techniques.

Gamification is often defined as the use of game design elements in non-game contexts [26], usually with the main goal of engaging and motivating users. Gamification is a complex phenomenon that involves various elements, such as game mechanics, dynamics, and components, as well as psychological theories and motivational frameworks. Therefore, the knowledge acquisition process for the MAGO-Ga ontology involves researching the existing ontologies applicable to describing gamification, and identifying the most relevant concepts and relationships.

The main identified and selected sources of knowledge on gamification are the following ontologies:

- OntoGamif [27],
- Gamification Domain Ontology (GaDO) [28].

ONTOGAMIF is one of the earliest ontologies for gamification [27]. This ontology is a modular ontology covering several sub-domains of the domain of gamification:

- 1) Core Gamification Concepts (generic game design elements and principles),

- 2) Organizational Concepts (context of the implementing organization),
- 3) Evaluation Concepts (metrics and assessment of gamification outcomes),
- 4) Psychological Concepts (theories of motivation, player types, user behaviour),
- 5) Ethical Concepts (ethical considerations of gamification, e.g. manipulation, privacy),
- 6) Risk Concepts (potential negative side-effects or risks of gamification), and
- 7) User-Related Concepts (profiles of target users/players).

OntoGamif was built by extracting definitions of gamification elements from a broad literature review. Each class of this ontology is annotated with its source reference. The ontology is grounded in the SUMO upper ontology to ensure alignment with higher-level concepts. A key motivation for OntoGamif was to establish an agreed-upon common representation of gamification knowledge, which can facilitate communication among stakeholders (researchers, designers, managers) and enable semantic reasoning about gamified systems.

In terms of contribution, OntoGamif provides a holistic semantic model of gamification, unifying concepts from game design, psychology, and implementation practice. Notably, the authors suggest it can help managers understand employee behaviour and needs through gamification's theoretical lens, indicating its relevance to enterprise contexts (e.g. workplace gamification). OntoGamif is publicly available: the authors released the ontology's OWL file and documentation via the Mendeley Data repository.

GaDO is an ontology for gamification in the domain of intelligent tutoring systems (ITS) [28]. The main purpose of this ontology was 'to support its application in intelligent tutoring systems.' This ontology is divided into two sub-domain ontologies: 1) GaDO Core, and 2) GaDO Full.

GaDO Core covers the gamification domain, including game design elements, motivational theories, and user profiles. GaDO Full extends GaDO Core by adding ITS-specific concepts, such as student models, domain models, and pedagogical components. The main contribution of GaDO is the integration of gamification with ITS, enabling semantic modelling of the use of gamification in educational contexts. GaDO is publicly available. The GaDO ontology covers the gamification domain, including game design elements, motivational theories, and user profiles.

GaDO ontologies are used in the process of engineering the GaTO ontology for gamification in the domain of intelligent tu-

toring systems (ITS) [28]. The main purpose of this ontology was ‘connecting gamification and intelligent tutoring systems concepts.’

The ontology maps specific game design elements to these target behaviour categories, grounding gamification design choices in pedagogical objectives. By formalizing such relationships, GaTO supports automated reasoning (e.g., an intelligent tutor can infer which game element to deploy to foster a certain behaviour) and interoperability of gamified tutoring systems. The authors evaluated GaTO using ontology quality metrics and demonstrated a use-case of an ontology-driven authoring tool for teachers. GaTO is also available publicly.

These ontologies aim to support personalized gamification – adapting game elements to different learners – and to guide the design of gamified learning activities in a systematic way.

OTHER ontologies have been found in the domain of gamification, but most are focused on specific aspects or applications of gamification, rather than providing a comprehensive model of the domain. Education has been a particularly active domain for gamification, and consequently many recent ontologies focus on modelling gamification in educational contexts. These include ontologies for integrating gamification with intelligent tutoring systems [28], gamified collaborative learning sessions [29], personalised or adaptive gamification in online learning platforms [30], profiling students in the context of gamified learning systems [31], modelling user profiles and activities in gamified systems [32], and game-based assessment [33]. However, these ontologies are not directly applicable to the domain of multi-agent systems and are therefore not considered in the knowledge acquisition process for the MAGO-Ga ontology. Moreover, these ontologies are focused on specific aspects of gamification, rather than providing a comprehensive model of the domain. Therefore, they are considered only as inspiration in the knowledge acquisition process for the MAGO-Ga ontology.

THE CONCEPTS of the above mentioned two ontologies revolve, naturally, around the central concept of gamification and gamified systems. Since gamification by definition heavily depends on game design elements, the two ontologies also include a wide range of concepts related to game design. Finally, both ontologies include concepts related to the users of gamified systems, which is also relevant to the MAGO-Ga ontology since the agents in the system are the users of the gamified system and the players that can be found within it.

Some basic game design elements are mentioned in these ontologies, such as game mechanics, game dynamics, and game components. In *OntoGamif*, gamification mechanics are defined as the functional components of a gameful system that provide actions and control mechanisms. Examples given include point systems, leaderboards, levels, and challenges. *GaDO* similarly defines Mechanic as

gamification mechanics

one of the three types of game design element. In GaDO, though, game mechanics are the rules or rewards that govern the gamified learning process. For modelling a gamified multi-agent system, mechanics may be vital in the context of describing what gamified incentives or constraints are present. Gamification dynamics are modelled as the individual's reactions or behaviours that arise in response to the gamified system's mechanics. In OntoGamif, dynamics are defined as the emergent reactions or behavioural outcomes that result from applying mechanics. For example, a mechanic like a leaderboard may create competition among players, wherefore competition is a dynamic, i.e. a user reaction or social interaction pattern. In GaDO, dynamics are not explicitly modelled, but the idea is similar: the behavioural outcomes of gamified interactions are of central importance. For a multiagent system, dynamics could describe how the agents behave in response to the gamified system's mechanics. Ultimately, gamification elements are modelled as components of a game, i.e. specific elements or tokens used in gamification. In OntoGamif, these are called gamification components and include elements like badges, points, leaderboards, and quests. In GaDO, the idea is similar: the concrete elements used in gamification are of great importance. For a multiagent system, components might describe the specific elements used in the gamified system. Usually there are a limited number of game components that are used most often, although the comprehensive set is much larger. The most well-known components include:

gamification dynamics

gamification elements

- Badge – a symbolic reward (usually an icon or title) given for accomplishing something, e.g. a 'Problem Solver' Badge after solving 100 puzzles.
- Point – a numeric score that accumulates, often for accomplishing tasks or reaching milestones, e.g. earning points for each completed task. Points can be used in different rules, such as to unlock new levels or to compete on leaderboards.
- Leaderboard – a ranked list of players by some score, often used to encourage competition and social comparison. For example, one player might try to outscore another player to reach the top of the leaderboard.
- Quest – a series of tasks or challenges that must be completed to earn a reward. For example, a player might be given a quest to complete 10 tasks within a certain time limit to earn a reward.
- Level – a progressive stage that agents can achieve, e.g. an agent might reach a new level after completing a certain number of tasks or achieving a certain score.

badge

point

leaderboard

quest

level

PERSONALITY modelling is one of the many interesting aspects of gamification and modelling a gamified system, since personality

traits and facets are known to influence the effectiveness of gamification techniques [34], [35], [36], [37]. For example, extraverts tend to enjoy competitive and social elements like leaderboards, whereas introverts might prefer unlockable content or single-player challenges. Conscientious individuals might respond well to achievement badges and clear goals, while those high in Openness might appreciate narrative and exploration mechanics.

The Big Five personality model, also known as the Five-Factor Model [25], also known as OCEAN for the five personality traits it includes (Openness, Conscientiousness, Extraversion, Agreeableness, Neuroticism), is a widely accepted framework for describing human personality facets. While neither OntoGamif nor GaTO explicitly provide a full Big Five trait ontology out of the box, OntoGamif’s user model is designed to accommodate personality aspects, and it can be extended to incorporate the Big Five traits and facets. However, OntoGamif does not list each trait as a separate class. GaTO itself doesn’t incorporate the Big Five, but it does incorporate the BrainHex player types as a form of gamer psychology modelling. BrainHex types (e.g. Seeker, Survivor, Daredevil) correlate with certain play styles, which are arguably related to personality traits (for example, BrainHex Socializer overlaps with being high in extraversion/agreeableness).

In the context of multi-agent systems, if our agents are simulations of humans or user personas, giving them personality profiles (even randomly assigned or scenario-specific) can lead to more diverse and realistic behaviour patterns. For example, in a multi-agent training simulation, we might want some agents to be competitive (high extraversion/low agreeableness) and others to be cooperative (high agreeableness) to see how they interact under a gamified incentive scheme.

Selecting the most relevant concepts from the ontologies, the following list of concepts is identified as relevant to the context of modelling gamification in the domain of multi-agent systems.

Concept name	Gamification mechanics
Description	A functional game design element that provides actions or control in a system, e.g. points, leaderboards, levels, challenges. Mechanics make up the functioning components of the game and allow the designer control over player actions.

Table 2.1: *Gamification mechanics* glossary entry

Concept name	Gamification dynamics
Description	The reactions and behaviours that emerge from gameplay as a response to mechanics. For example, competition among players arises from certain mechanics such as leaderboards, while team-based challenges might result in cooperation. Dynamics are individuals' reactions as a response to implemented mechanics.

Table 2.2: *Gamification dynamics* glossary entry

Concept name	Game component
Description	A specific gamification element or token in the system, e.g. a badge awarded for achievements, a ranked list of players by score, a numerical score that can be earned, etc. These are the visible artefacts of gamification given to or seen by players.

Table 2.3: *Game component* glossary entry

Concept name	Player
Description	The actor or agent who participates in the gamified system. Players are classified by type and can have a profile.

Table 2.4: *Player* glossary entry

Concept name	Player type
Description	A categorisation of player/users by motivational or behavioural profile. OntoGamif's user typology includes 10 types combining Bartle's and Hexad models. GaDO includes player types from the BrainHex model. A player type describes how a player tends to engage with the system (e.g. achievers seek competency and challenges, socialisers seek social interaction etc.).

Table 2.5: *Player type* glossary entry

Concept name	Personality trait
Description	A personality dimension of an actor (e.g. user or agent) defined by the Five Factor model: openness, conscientiousness, extraversion, agreeableness, neuroticism.

Table 2.6: *Personality trait* glossary entry

Concept name	Gamification objective
Description	The goal or purpose of implementing gamification in a system - essentially why we are gamifying. For example: increase motivation, improve productivity, or encourage desired behaviour.

Table 2.7: *Gamification objective* glossary entry

Concept name	Target behaviour
Description	A specific user behaviour to encourage or change via gamification. GaTO defines target behaviour categories like participation, performance, exploration, enjoyment, effectiveness, and competition. For example, ‘increase forum posts’ might be a target behaviour under participation.

Table 2.8: *Target behaviour* glossary entry

Concept name	Engagement loop
Description	A cyclical interaction loop consisting of motivation → action → feedback, which sustains engagement over time. In GaTO, an engagement loop is a type of activity loop, and it ties together game elements providing motivation, the player’s action, and feedback (a game mechanic delivering response). Essentially, it models how the system continually engages the player.

Table 2.9: *Engagement loop* glossary entry

3

Conceptualisation

Based on the knowledge acquired in the previous chapter, the next step is to transform the collected knowledge into a structured and meaningful model. During this phase, the key concepts, relationships, attributes, and rules of the domain are identified and organized to reflect the domain's structure. The goal is to create a clear and coherent conceptual model that captures the domain's essential elements without formalizing them into specific ontology languages. The conceptual model produced in this phase is not yet formalized but provides a foundational blueprint for the subsequent formalization and implementation phases of the ontology development process.

Several concepts have been identified as relevant to the context of modelling gamification in the domain of multi-agent systems. The visual representation of the conceptual model is based on the concepts of three ontologies. The visual representation of the conceptual model is an extension of the GIVEn ontology. The concepts that are not going to be included in the final ontology are absent from the visual conceptual model.

The selection of the concepts for the MAGO-Ga ontology is based on the intended use of the ontology. The ontology is intended to model gamified multi-agent systems, wherefore the concepts related to gamification are carefully chosen. The concepts related to the agents and their personalities are also included, as they are essential for modelling the interactions between the agents and the gamification elements.

The following list of tables provides a detailed description of the selected concepts, their definitions, and their intended use in the ontology.

© see figure 3.1

© see tables 3.1 – 3.20

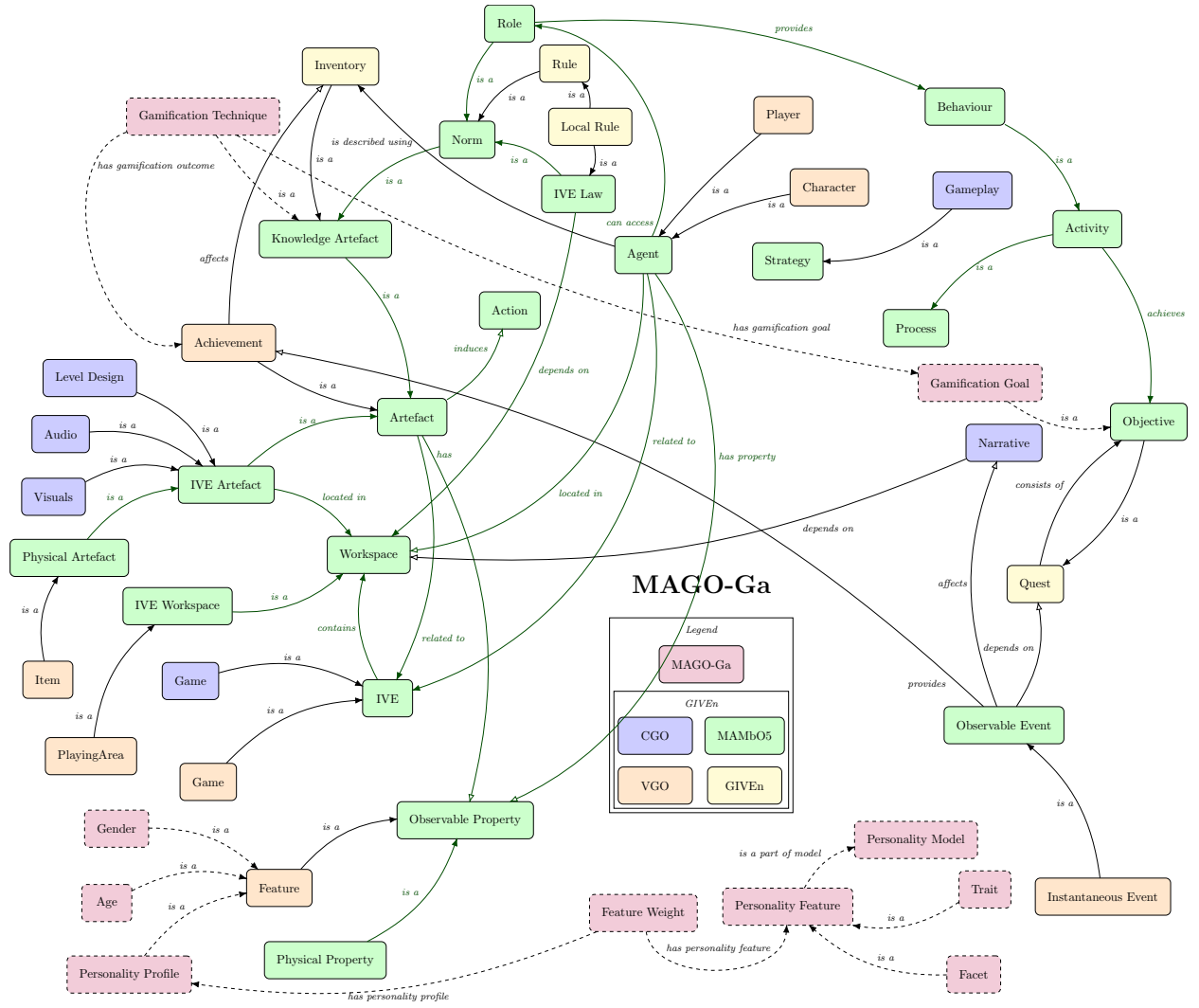


Figure 3.1: Concepts and their relationships in the MAGO-Ga ontology

Concept name	Gamification technique
Definition	A method or mechanism used to introduce game elements or principles into a non-game context for agents
Description	This class represents a specific gamification method applied in the multiagent environment. A gamification technique can include elements like point systems, badges, challenges, or other game-like incentives tailored for artificial agents. It is considered a knowledge artefact in the ontology, meaning it is a defined strategy or design that can be implemented to influence agent behaviour and motivation.

Table 3.1: *Gamification technique* concept details

Concept name	Gamification goal
Definition	The objective that must be achieved in order to fulfil a gamification technique
Description	A gamification goal is the quantified objective of the agent that the agent must achieve in order to fulfil a gamification technique. For example, a gamification goal might be to increase the number of interactions between agents, to improve the quality of their interactions, or to achieve a specific task such as producing a certain amount of a certain resource.

Table 3.2: *Gamification goal* concept details

Concept name	Personality model
Definition	An abstract framework defining a set of personality features and their relationships
Description	A personality model provides the theoretical structure for organising personality traits in the ontology. For example, the Five-Factor Model (Big Five) is a personality model that includes traits like Openness and Extraversion. In this ontology, the personality model class encompasses such frameworks and specifies which personality features (traits or facets for the Five-Factor Model) belong to the model. It allows the system to interpret an agent's personality profile according to a well-defined set of traits.

Table 3.3: *Personality model* concept details

Concept name	Personality feature
Definition	A characteristic of personality, such as a trait or a facet, that can be used to describe personality of an agent
Description	A personality feature represents here any measurable aspect of personality that an agent can have defined within a personality model. It serves as a general category encompassing broad traits (like Extraversion or Conscientiousness) and more specific facets. Personality features form the building blocks of personality profiles and are defined within a personality model. By capturing these features, the system can adapt gamification elements to match individual personality differences.

Table 3.4: *Personality feature* concept details

Concept name	Trait
Definition	A major personality trait representing a broad dimension of personality in the Five-Factor Model
Description	Trait refers to a high-level personality characteristic of an agent. Traits are typically the primary factors in a personality model, such as the Big Five traits (Openness, Conscientiousness, Extraversion, Agreeableness, Neuroticism). Each trait encompasses several facets and describes consistent patterns in an agent's behaviour or disposition. In this ontology, traits help structure the agent's personality profile, allowing broad-strokes customisation of gamification strategies based on an agent's dominant personality characteristics.

Table 3.5: *Trait* concept details

Concept name	Facet
Definition	An aspect of a personality trait
Description	In personality modelling, facets are finer-grained characteristics that make up a larger trait. They provide a detailed breakdown of personality features. For example, within the trait of Extraversion, facets might include sociability or assertiveness. Facets in the ontology allow for more granular distinctions in an agent's personality profile, supporting more precise gamification personalisation and determining the strength of the effect of gamification techniques on agents.

Table 3.6: *Facet* concept details

Concept name	Personality profile
Definition	A collection of personality features and their associated weights that characterises a specific agent
Description	This class represents the personalised set of personality attributes for an individual agent. A personality profile consists of multiple personality features (e.g. traits or facets), each with an associated feature weight, effectively mapping out the agent's personality. It is used by the framework to tailor gamification techniques to the agent. For instance, an agent's profile may indicate high Openness and low Neuroticism, which can guide the selection of suitable motivational strategies. Furthermore, the personality profile can be used to determine the effect of gamification techniques on agents, as different techniques might have different effects on agents with different personality profiles.

Table 3.7: *Personality profile* concept details

Concept name	Feature weight
Definition	A numeric weighting assigned to a personality feature, indicating its intensity
Description	This class represents the quantified level of a particular personality feature in an agent's profile. Each feature weight links a personality feature (specifically a facet in the developed framework) to a value (often a floating-point number) that reflects how strongly the agent exhibits that feature. It is used to create a detailed personality profile, enabling the gamification framework to adjust strategies based on individual differences.

Table 3.8: *Feature weight* concept details

Concept name	Age
Definition	A demographic attribute representing how old an agent (or user) is
Description	In this ontology, age is modelled as a type of feature associated with an agent. It denotes the temporal duration of an agent's existence (for example, in years) or an age category. Age can be used to tailor gamification elements, as different age groups might respond to game mechanics differently.

Table 3.9: *Age* concept details

Concept name	Gender
Definition	A personal attribute representing the gender of an agent
Description	Gender is modelled as a feature of an agent. It represents an agent's gender identity (such as male, female, or other categories) and is included as a part of the agent's profile. This attribute can be relevant in contexts where gender might influence preferences or behaviour, thereby informing how gamification is tailored to different agents.

Table 3.10: *Gender* concept details

Concept name	defines feature
Definition	Indicates that a personality model includes a given personality feature as part of its defined set
Description	This object property is the inverse of "is a part of model". It links a personality model to a personality feature, meaning the feature is defined by or within that model. For example, the Five-Factor Model defines features such as Openness and Extraversion. Using "defines feature", it can be designated which traits or facets a particular personality model encompasses.
Domain	Personality Model
Range	Personality Feature

Table 3.11: *defines feature* concept details

Concept name	has event object
Definition	Links an observable event to a physical artefact that is involved in that event
Description	This property associates an event occurring in the environment with the object (artefact) that the event involves. In a multiagent or virtual environment, an "Observable Event" might have a "Physical Artefact" as its object. For instance, a "pick-up" event has the picked item as the event object. "Has event object" thus helps describe the context of events by specifying what object is affected or used.
Domain	Observable Event
Range	Physical Artefact

Table 3.12: *has event object* concept details

Concept name	has gamification goal
Definition	Associates a gamification technique with the gamification goal it is intended to fulfil
Description	This property ties a gamification technique to its purpose or intended objective. The gamification goal is an abstract target (such as improving collaboration or increasing engagement) that the player must achieve in order to fulfil the gamification technique. By linking techniques to goals, the ontology clarifies why each gamification method is applied.
Domain	Gamification Technique
Range	Gamification Goal

Table 3.13: *has gamification goal* concept details

Concept name	has gamification outcome
Definition	Associates a gamification technique with an outcome event that results from its application
Description	When a gamification technique is applied, it produces a certain outcome or reward event. The "has gamification outcome" property connects the technique to that resulting "Gain Event". For instance, a points-award technique will have a point-gain event as its outcome, whereas a badge-award technique will have a badge-gain event as its outcome. This link allows the system to know what immediate effect or reward is generated by the gamification mechanism in the environment.
Domain	Gamification Technique
Range	Gain Event

Table 3.14: *has gamification outcome* concept details

Concept name	has observable property
Definition	Links an agent to a property of that agent that can be observed in the environment
Description	This property denotes that an agent has a certain attribute which is observable by other agents or the system. An observable property might be something like the agent's current score, location, or status. By using "has observable property", the ontology can represent which aspects of an agent's state are externally visible, enabling agents or systems to respond to those observable attributes.
Domain	Agent
Range	Observable Property

Table 3.15: *has observable property* concept details

Concept name	has personality feature
Definition	Connects a feature weight to the specific personality feature it represents
Description	This object property links a feature weight instance to the personality feature (trait or facet) that the weight measures. Essentially, it specifies which personality characteristic a given numeric weight corresponds to in an agent's profile. For example, a weight value might be connected to the feature "Extraversion", indicating that the number quantifies the agent's level of Extraversion.
Domain	Feature Weight
Range	Personality Feature

Table 3.16: *has personality feature* concept details

Concept name	has personality profile
Definition	Associates an agent, a gamification technique, or a feature weight) with a particular personality profile
Description	This functional property is primarily used to link an agent or a gamification technique to their personality profile. It can also connect a feature weight to the personality profile it belongs to, grouping individual trait measurements under a single profile. By using "has personality profile", the ontology identifies which profile describes an agent's personality, enabling the system to treat that set of traits as a unit when adapting gamification strategies.
Domain	Agent or Feature Weight
Range	Personality Profile

Table 3.17: *has personality profile* concept details

Concept name	has weight value
Definition	Specifies the numeric value of a feature weight, typically as a floating-point number
Description	This data property holds the quantitative value for a feature weight entry. The value is usually a float that indicates how strongly an agent possesses a particular personality feature. Through "has weight value", each trait in an agent's personality profile is given a magnitude, which can be used by the gamification logic to adjust game elements (such as difficulty or rewards) according to the agent's personality.
Domain	Feature Weight
Range	Float

Table 3.18: *has weight value* concept details

Concept name	is a part of model
Definition	Indicates that a personality feature is a component of a larger personality model
Description	This property represents the relationship where a personality feature (e.g. a trait or a facet) belongs to a personality model. In other words, it declares that a given trait is part of the structure of a particular personality framework. For example, 'Extraversion is a part of the Five-Factor model.' "Is a part of model" is the inverse of "defines feature" and helps organise features under their respective models.
Domain	Personality Feature
Range	Personality Model

Table 3.19: *is a part of model* concept details

Concept name	has name
Definition	Assigns a textual name to an entity in the ontology
Description	"Has name" is a data property used to provide a human-readable name or title for an individual or entity in the ontology. It serves as an identifier that is easier to read and communicate than an IRI. For example, a particular gamification technique instance could be given a name like "Price Optimiser" using this property. This makes it convenient to reference entities in interfaces or documentation.
Range	String

Table 3.20: *has name* concept details

4

Formalisation

The formalisation phase in ontology engineering focuses on applying formal semantics to the conceptual model of a domain. This phase aims to create an ontology that is precise, unambiguous, and capable of supporting complex reasoning. By using a formal language, the ontology's structure, relationships, and constraints become systematically interpretable by both humans and machines.

The formalisation of the MAGO-Ga ontology is based on the Web Ontology Language (OWL). OWL is a family of knowledge representation languages for authoring ontologies. OWL ontologies are based on the Resource Description Framework (RDF) and are designed to be machine-readable and machine-interpretable. Even so, MAGO-Ga ontology is given a human-readable documentation in this deliverable, using Turtle syntax serialization, as well as a machine-readable formalisation in the OWL format.

© see [MAGO-Ga Turtle Serialization](#)

5

Integration

The integration phase of this ontology engineering methodology focuses on harmonizing the developed ontology with other relevant ontologies or knowledge sources to create a unified knowledge ecosystem. This phase enriches the ontology's content, enhancing its semantic reach and improving its reusability across various applications.

Connecting with external ontologies supports data consistency, reduces redundancy, and ensures that the developed ontology can communicate effectively within broader systems, ranging from semantic web applications to intelligent multi-agent frameworks.

MAGO-Ga ontology is designed as an extension of the GIVEn ontology. The GIVEn ontology is a domain ontology for the domain of games as multiagent intelligent virtual environments. The GIVEn ontology is designed to model the elements of an intelligent virtual environment and their interactions, enhanced with organisation-related elements. The MAGO-Ga ontology extends the GIVEn ontology with the concepts related to gamification and the agent personality profiles.

Due to the nature of the aim and intended use of the MAGO-Ga ontology, no further integration with other ontologies is foreseen as necessary. Nonetheless, the MAGO-Ga ontology is designed to be compatible with other ontologies, should the need arise to extend its scope in the future. In particular, MAGO-Ga ontology may be extended, for example, with the concepts of the MAGO-Ag ontology, which is designed to model the implementable agents and their interactions in a multi-agent system. The visualised conceptual model already shows how some other ontologies are integrated with the MAGO-Ga ontology. Depending on the implementation needs in the future, the MAGO-Ga ontology may be extended or further established with the concepts from other ontologies.

© see figure 3.1

6

Implementation and Evaluation

6.1 Implementation

The implementation phase of the ontology engineering process involves creating a concrete representation of the ontology in a specific format or tool. This phase ensures that the ontology is not only theoretically sound but also practically usable. The implementation phase may include tasks such as developing software tools for managing and querying the ontology, creating user interfaces for accessing the ontology's knowledge, and integrating the ontology with existing systems or applications.

The MAGO-Ga ontology is implemented using the Protégé tool for engineering ontologies. The implemented ontology is available on the GitHub repository of this project and is OWL/XML serialized in appendix [MAGO-Ga OWL/XML Serialization](#).

 <https://github.com/AILab-FOI/MAGO/blob/main/Deliverables/Phase%202/Implementation/MAGO-Ga.rdf>

6.2 Evaluation

The MAGO-Ga ontology was evaluated by the domain experts from the host institution using the applicable interview and review approaches. Furthermore, the ontology was tested against the specification document.

© see [Specification](#)

Part III

The Gamification of an Intelligent Virtual Environment Framework MAGO-Ga

1

Introduction

The MAGO-Ag framework was planned to be an artificial agent implementation framework that would make it possible for agents to establish interaction with gamification techniques, i.e. for gamification techniques to influence the agent behaviour. The framework was envisioned to provide gamification knowledge to agents via the developed ontologies (primarily MAGO-Ga ontology) as the relevant knowledge models or knowledge bases.

The developed system, that implements the MAGO-Ag framework, is called GEAR – Gamified Emergent Agent Relations. This chapter describes GEAR and how it is implemented.

 <https://github.com/AILab-FOI/GEAR>

The main premise of GEAR is based on the recipeWorld model [38]. The agents in the system are service providers and consumers, and service consumers have a list of services they need to consume, called a recipe. The service providers offer a subset of all the services available in the system. The service consumers move from one service provider to another based on their respective recipes, their required services, the availability of the service providers etc. This commotion enables connections to be formed based on the sequence of completed recipe elements.

The original recipeWorld model does not include gamification elements. In GEAR, the gamification techniques are used to influence agent behaviour within the context of the implemented system. Personality profiles of the agents are based on the Five-Factor Model of personality traits, i.e. their associated facets [25]. The compatibility between gamification techniques and agents is calculated as a distance between the personality vectors of the subject agent and the gamification techniques.

The approach developed and presented here seeks to provide a system of artificial agents whose behaviour is influenced by gamification techniques, based on their psychological profile using personality facets. The agents are psychologically profiled using the Big Five personality traits and their associated facets. The gamification techniques are associated with personality profiles, enabling the effective tailoring of agent behaviour. Furthermore, agent personality affects some other choices the agents make, such as the prices they are willing to accept.

2

Formalisation and Implementation

This chapter presents the formalisation of the gamification elements and their interactions with the agents in the system, as well as the implementation of the proposed approach. The fundamental concepts used here are also present in the MAGO-Ga ontology, but the formalisation presented here is of a different nature and aims at describing the proposed approach in a more mathematical way.

The following section provides a formalisation of the gamification elements and their interactions with the agents in the system. The content provided here was submitted as a manuscript to the EUMAS conference on multiagent systems.

2.1 Theoretical Foundation and Formalisation

The foundational elements of this system are agents, personality traits and facets, and gamification techniques. Agents are described using personality traits, which are in turn described using personality facets. Gamification techniques are related to specific personality profiles also described using personality facets. The compatibility between gamification techniques and agents is determined based on the compatibility between their personality profiles. In this context, compatibility denotes the degree to which a gamification technique is suitable for an agent, based on their personality, i.e. how effective is a gamification technique in influencing the behaviour of an agent with a specific personality. Therefore, some agent are more susceptible to the influence of a gamification technique, while others are not.

Let $\mathcal{A} = \{\alpha_1, \alpha_2, \dots, \alpha_n\}$ be the set of all agents, where each $\alpha_i \in \mathcal{A}$ represents an individual agent.

agents

The set of personality facets constituting a personality profile is chosen based on the research of McCrae and Costa [25]. The five personality traits in their Five-Factor Model are openness, conscientiousness, extraversion, agreeableness, and neuroticism. Each of these traits is further divided into six facets that influence each trait:

- Openness: Fantasy, Aesthetics, Feelings, Actions, Ideas, Values;
- Conscientiousness: Competence, Order, Dutifulness, Achievement striving, Self-Discipline, Deliberation;

- Extraversion: Warmth, Gregariousness, Assertiveness, Activity, Excitement seeking, Positive emotions;
- Agreeableness: Trust, Straightforwardness, Altruism, Compliance, Modesty, Tender-mindedness;
- Neuroticism: Anxiety, Angry hostility, Depression, Self-Consciousness, Impulsiveness, Vulnerability.

For example, an agent with a high score in the ‘Warmth’ facet of the ‘Extraversion’ trait is more likely to be influenced by gamification techniques that rely on social interaction. On the other hand, an agent with a high score in the ‘Anxiety’ facet of the ‘Neuroticism’ trait is more likely to be influenced by gamification techniques that rely on individual performance.

Let $\Phi = \{\phi_1, \phi_2, \dots, \phi_{30}\}$ be the set of all personality facets, where each $\phi_i \in \Phi$ corresponds to a distinct facet (e.g., *Feelings* is ϕ_3 , *Vulnerability* is ϕ_{30}).

personality facets

Let $\Psi = \{\vec{\psi}_1, \vec{\psi}_2, \dots, \vec{\psi}_m\}$ be the set of all personality profiles, where each $\vec{\psi}_i \in \Psi$ represents a unique personality profile. Personality profiles are mined from empirical data and are not necessarily unique to each agent, so m need not equal the number of agents n . A personality profile $\vec{\psi}_i$ represents an individual personality instance as a 30-dimensional real-valued vector (2.1), where each component $x_{\phi_i} \in [0, 1]$ indicates the intensity of the corresponding facet ϕ_i .

personality profiles

$$\vec{\psi}_i = [x_{\phi_1}, x_{\phi_2}, \dots, x_{\phi_{30}}] \in [0, 1]^{30} \quad (2.1)$$

This relationship can be observed in MAGO-Ga ontology as the **Feature weight** class. The value of the feature weight is represented by the **has weight** value property, and the personality feature it corresponds to is related via the **has personality feature** property.

⊗ see table 3.8

⊗ see table 3.18

⊗ see table 3.16

Each agent $\alpha_i \in \mathcal{A}$ is given a personality profile $\vec{\psi}_i \in [0, 1]^{30}$. However, personality is not a simple concept to model. To account for additional factors, e.g. experience or rebelliousness, a scalar value $\chi_i \in [0, 2]$ is added. Thus, the overall personality representation of an agent i , i.e. p_i , is defined as a tuple (2.2).

$$p_i = (\chi_i, \vec{\psi}_i) \quad (2.2)$$

The personality modifier χ_i acts as a scaling factor that can either reduce (< 1) or enhance (> 1) the compatibility score. For the sake of simplicity, we assume that the personality profile of an agent is fixed and does not change over time.

personality modifier

Gamification techniques are various approaches used in a gamified system to influence the behaviour of users. Certain gamification techniques have a specific degree of compatibility or influence over specific personality profiles. Therefore, it is proposed that gamification techniques can be related to specific personality profiles, also described using personality facets. Additionally, each

gamification technique is given a name and is described using two more features: a goal and a reward. The goal represents the condition that must be met to earn the reward, while the reward is the item or benefit that is awarded when the goal is achieved.

Let $\Gamma = \{\gamma_1, \gamma_2, \dots, \gamma_r\}$ be the set of all gamification techniques, where each $\gamma_i \in \Gamma$ is defined as a tuple (2.3).

gamification techniques

$$\gamma_i = (\text{name}_i, \text{goal}_i, \text{reward}_i, \vec{\psi}_i) \quad (2.3)$$

Here, $\vec{\psi}_i \in \Psi$ denotes the personality profile that the technique γ_i is most likely to influence. Therefore, $\gamma_i \in \mathcal{N} \times \mathcal{G} \times \mathcal{R} \times \Psi$, where $\mathcal{N}$ is the set of all possible names (e.g., strings or labels), $\mathcal{G}$ is the space of goals, and $\mathcal{R}$ is the space of rewards.

It should be noted here that this tuple can be observed in the MAGO-Ga ontology as the `Gamification technique` class. The name of the gamification technique is represented by the `has name` property, the goal by the `has gamification goal` property, the reward by the `has gamification outcome` property, and the personality profile by the `has personality profile` property.

⊙ see table 3.1

⊙ see table 3.20

⊙ see table 3.13

⊙ see table 3.14

⊙ see table 3.17

For example, the technique $\gamma_1 = (\text{Leaderboard}, g_1, r_1, \vec{\psi}_1)$ might specify that achieving the top 3 rank (goal g_1) grants a badge (reward r_1), and is most influential on individuals with high competitiveness ($\vec{\psi}_1$). Personality profile $\vec{\psi}_1$ may already be defined in the set of personality profiles Ψ and associated with some other gamification techniques or agents, or it may be a new profile that is added to the set.

The compatibility between a gamification technique γ_i and an agent α_j is determined based on the Euclidean similarity of their personality profiles $\vec{\psi}_i$ and $\vec{\psi}_j$, adjusted by the agent's scalar personality modifier $\chi_j \in [0, 1]$. The compatibility is calculated using the Euclidean distance between the two personality profiles, as shown in (2.4).

compatibility

Euclidean similarity

$$c(\gamma_i, \alpha_j) = \min \left(1, \chi_j \cdot \left(1 - \frac{\|\vec{\psi}_i - \vec{\psi}_j\|_2}{\sqrt{30}} \right) \right), \quad c : \Gamma \times \mathcal{A} \rightarrow [0, 1] \quad (2.4)$$

Here, $\|\cdot\|$ denotes the Euclidean norm, and the denominator $\sqrt{30}$ normalises the distance so that the similarity term lies in $[0, 1]$. $\chi_j \in [0, 1]$ is an agent's personality modifier. If $\chi_j > 1$, compatibility is amplified; if $\chi_j < 1$, it is suppressed. The result is clamped to ensure $c(\gamma_i, \alpha_j) \in [0, 1]$. This yields a compatibility score $c(\gamma_i, \alpha_j) \in [0, 1]$, where 1 indicates maximum compatibility and 0 indicates no compatibility.

2.2 Model Implementation

The implemented system to showcase the approach proposed in this paper is based on the recipeWorld model [38]. The agents in the system are service providers and consumers, and the gamification techniques are used to influence their behaviour in that

context. The personality profiles of the agents are based on the Five-Factor Model of personality traits and their associated facets [25]. The compatibility between gamification techniques and agents is calculated using the formula presented in (2.4).

The proposed system is implemented in Python, using the NumPy library for vector and matrix operations. The implementation includes functions for calculating the compatibility between gamification techniques and agents, as well as for simulating the behaviour of agents in a gamified system. The implementation is designed to be modular and extensible, allowing for the addition of new gamification techniques, agents, and personality profiles.

Agents of the proposed system are implemented using the SPADE framework [39], [40], [41]. Each agent is extended using a personality module that includes at least the personality profile. The same personality profile class is used to instantiate personality profiles for agents and gamification techniques.

Some instances of how an agent behaves are influenced by the agent's personality. For example, the value of an offer an agent is observing is influenced by the agent's personality. The value of an offer is calculated based on the offered service's price and duration, influenced by the number of awards showcased by the service provider, as well as the agent's personality traits related to price sensitivity (self-discipline), quality focus (dutifulness), badge appreciation (values), and risk tolerance (excitement seeking). The implementation of the offer value calculation is shown in listing 2.1.

<https://github.com/AILab-FOI/GEAR>

```

1 price_sensitivity = agent.personality.get_facet_score("Self-Discipline")
2 quality_focus = agent.personality.get_facet_score("Dutifulness")
3 badge_appreciation = agent.personality.get_facet_score("Values")
4 risk_tolerance = agent.personality.get_facet_score("Excitement seeking")
5
6 offer_values = []
7 for offer in offers:
8     ...
9     price_value = service.price
10    duration_value = service.duration
11    awards_value = awards
12
13    price_component = price_value * (0.5 + price_sensitivity * 0.5)
14    duration_component = -duration_value * (quality_focus * 0.5)
15    badge_component = -awards_value * (badge_appreciation * 10)
16    risk_component = (random.random() * 2 - 1) * risk_tolerance * 5
17
18    weighted_value = (
19        price_component + duration_component + badge_component + risk_component)
20    offer_values.append(
21        {"provider": provider, "service": service, "value": weighted_value})

```

Listing 2.1: Determining value of an offer based on personality

As mentioned earlier, compatibility between gamification techniques and agents is calculated using the formula presented in (2.4). The implementation of the body of the function performing the compatibility calculation is shown in listing 2.2.

```

1 technique_vector = self.personality_profile.get_personality_vector()
2 agent_vector = agent.personality.get_personality_vector()
3 agent_personality_modifier = agent.personality.get_personality_descriptor().get("personality modifier")
4
5 distance = np.linalg.norm(technique_vector - agent_vector)
6 max_distance = np.sqrt(30)
7 compatibility = min(1,
8     agent_personality_modifier * (1 - (distance / max_distance)))
9
10 return compatibility

```

Listing 2.2: Compatibility calculation wherein self is an instance of the gamification technique class and agent_personality is an instance of the personality profile class

An individual personality profile is implemented as a 30-dimen-

sional NumPy vector, where each dimension represents the score of a personality facet. The personality profile class includes methods for getting and setting the scores of individual facets, as well as for getting and setting the scores of all facets at once. A personality profile example, that of a personality describable as a *creative innovator*, is shown in listing 2.3.

```

1 creative_innovator = PersonalityProfile(np.array([
2     0.90, 0.85, 0.80, 0.85, 0.90, 0.88,    # Openness
3     0.70, 0.60, 0.65, 0.80, 0.70, 0.60,    # Conscientiousness
4     0.60, 0.65, 0.70, 0.75, 0.80, 0.70,    # Extraversion
5     0.70, 0.65, 0.70, 0.68, 0.60, 0.75,    # Agreeableness
6     0.30, 0.25, 0.20, 0.30, 0.40, 0.30    # Neuroticism
7 ]))

```

Listing 2.3: Personality profile

In addition to calculating a general compatibility of a gamification technique to a given agent based solely on their associated personality profile, the compatibility is also affected by the reward that is offered for the service. The reward compatibility is calculated based on the type of reward and the personality of the agent. For example, an agent with a high score in the *Achievement striving* facet of the *Conscientiousness* trait is more likely to be influenced by a gamification technique that offers a reward. If the agent has a high score in *Assertiveness*, it may be derived that it is oriented towards building status and is therefore again likely to be influenced by a gamification technique that offers a reward. These two personalities are currently set up to value badges and trophy rewards the most. A part of the body of the function performing the reward compatibility calculation is shown in listing 2.4.

```

1 achievement_drive = agent.personality.get_facet_score("Achievement striving")
2 status_orientation = agent.personality.get_facet_score("Assertiveness")
3 novelty_seeking = agent.personality.get_facet_score("Actions")
4
5 reward_compatibility = 0.5
6
7 reward_name = reward_item.name.lower()
8
9 if "trophy" in reward_name or "badge" in reward_name:
10     trophy_factor = 0.7 * achievement_drive + 0.3 * status_orientation
11     reward_compatibility = 0.6 + (trophy_factor * 0.4)
12 else:
13     other_factor = 0.6 * novelty_seeking + 0.4 * achievement_drive
14     reward_compatibility = 0.4 + (other_factor * 0.4)
15 return reward_compatibility

```

Listing 2.4: Reward compatibility calculation

Ultimately, the final compatibility between a gamification technique and an agent is calculated as a weighted average of the general compatibility and the reward compatibility, as shown in listing 2.5. In the particular example shown in listing 2.5, the general compatibility is weighted more heavily than the reward compatibility, as the general compatibility is a better indicator of how well a gamification technique will work for an agent.

```

1 technique_compatibility = self.calculate_compatibility(agent.personality)
2 reward_compatibility = self.calculate_reward_compatibility(agent, self.reward_item)
3
4 combined_compatibility = (technique_compatibility * 0.7) + (reward_compatibility * 0.3)

```

Listing 2.5: Combined compatibility calculation

3

Case Study: The Recipe World

This chapter presents more details on a case study that showcases the proposed approach. The case study is based on the recipeWorld model [38]. The agents in the system are service providers and consumers, and the gamification techniques are used to influence their behaviour in that context. The personality profiles of the agents are based on the Five-Factor Model of personality traits and their associated facets [25]. The compatibility between gamification techniques and agents is calculated using the formula presented in (2.4).

The implemented system is based on the recipeWorld model [38], which is ‘an agent-based model that simulates the emergence of networks out of a decentralized autonomous interaction.’ The main types of agents in the recipeWorld can be abstracted to the two types of agents present in the implemented system: service providers and service consumers. Service providers offer services to service consumers, who in turn request services from the providers. A list of services required by the consumers is defined as a recipe, and the consumers move from one provider to another based on their respective recipes and the required services. Not all services need to be provided by a single agent, and agents can specialise in providing one or more services. Due to this interaction, a social network emerges.

At the current state of the implemented system, the services that can be found in the system are defined using three attributes: a name, a price, and a duration. The name is a unique identifier for the service, the price is the baseline amount of money the service provider requires to offer the service, and the duration is the baseline amount of time the service provider requires to complete the service. The services are offered by service providers. On the side of service consumers, the services are aggregated into recipes, which are lists of services required by the consumers. The consumers then request services from the providers based on their respective recipes. A recipe can be considered a list of tasks that need to be completed in order to reach an objective – a finished recipe.

The communication between service consumers and service providers, which is the main interaction in the system, is shown in Fig.

3.1. While the service consumer agent has a recipe that is not yet completed, they will call for proposals from the service providers that offer the service required by the current recipe element. The service providers that offer the required service will then send a proposal to the service consumer, which includes the price and duration of the service. The service consumer will then select the best offer based on the price and duration of the service and number of rewards received by the service provider agent, and send a request to the selected service provider. The service provider will then either accept or reject the request. This decision is based on the service provider's capacity to provide the service, e.g. if the service provider is currently busy with another service request, they will reject the incoming request. If the request is accepted, the service provider will start providing the service, and the service consumer will wait for the service to be completed. Once the service is completed, the service consumer will move on to the next element in their recipe.

This communication flow, depicted in the sequence diagram, is controlled by a finite-state machine behaviour. This finite-state machine controls the behaviour of the service consumer agent.

Should a service required by the service consumer not be provided by any of the known service providers in the system, the service consumer will look for new providers. If no new service providers are found, the service consumer will stop searching for new providers and the recipe will be considered failed.

Service providers are designated specific services they offer when they are created, as shown in listing 3.1. The services are defined using the three attributes mentioned earlier. Service providers can offer multiple services, and a service can be offered by multiple service providers.

see figure 3.1

see figure 3.2

```

1 provider2.services = [Service("A", 8, 4), Service("C", 20, 2)]
2
3 ServiceProviderAgent{
4   "provider2@localhost",
5   "password",
6   provider2.services,
7   personality={
8     "personality profile": personality_profiles.creative_innovator
9   },
10 }
```

Listing 3.1: Service provider agent creation

If a service provider is faced with a gamification technique while it is making an offer to be sent as a proposal to a service consumer, the gamification technique may be applied to the offer. In the case of a gamification technique of type *service milestone*, which is instantiated as shown in listing 3.2, the offer is not necessarily changed, as per the definition of the gamification technique. In particular, this technique rewards the agent with a trophy for reaching a certain number of services provided. Based on the defined personality profile of the *service milestone* gamification technique, shown in listing 3.3, it is most effective on agents with a high conscientiousness trait. The personality profile of the service provider agent created

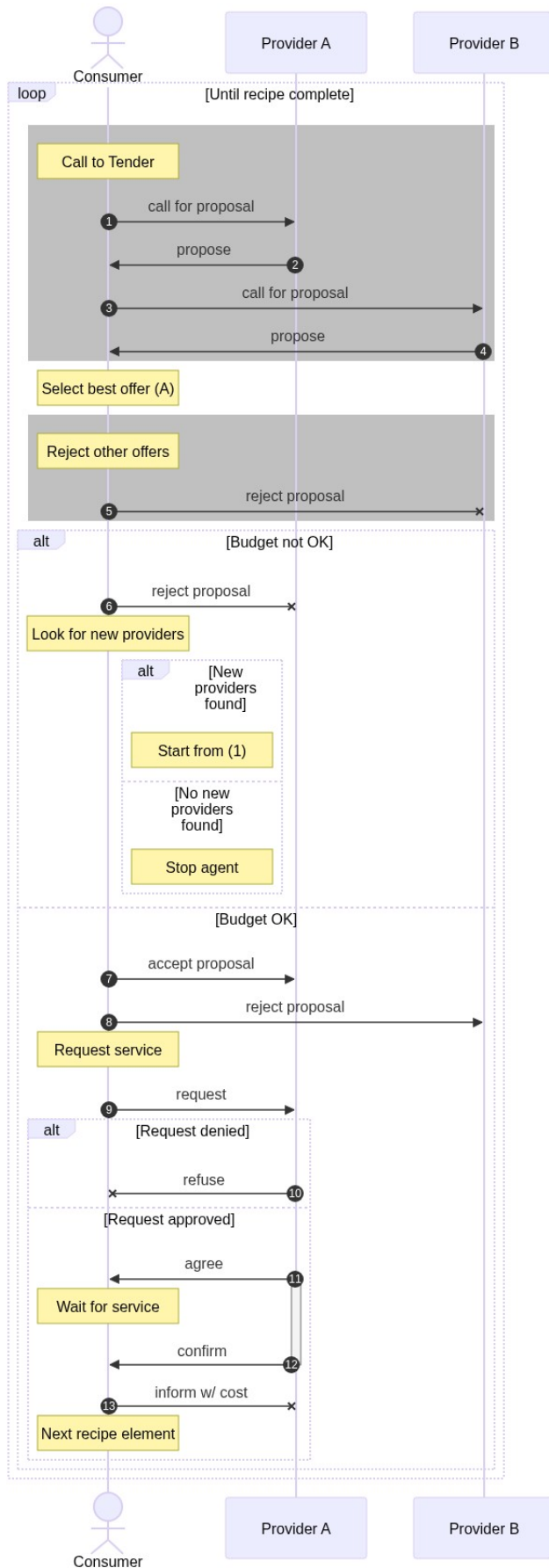


Figure 3.1: Sequence diagram of the service consumer agent

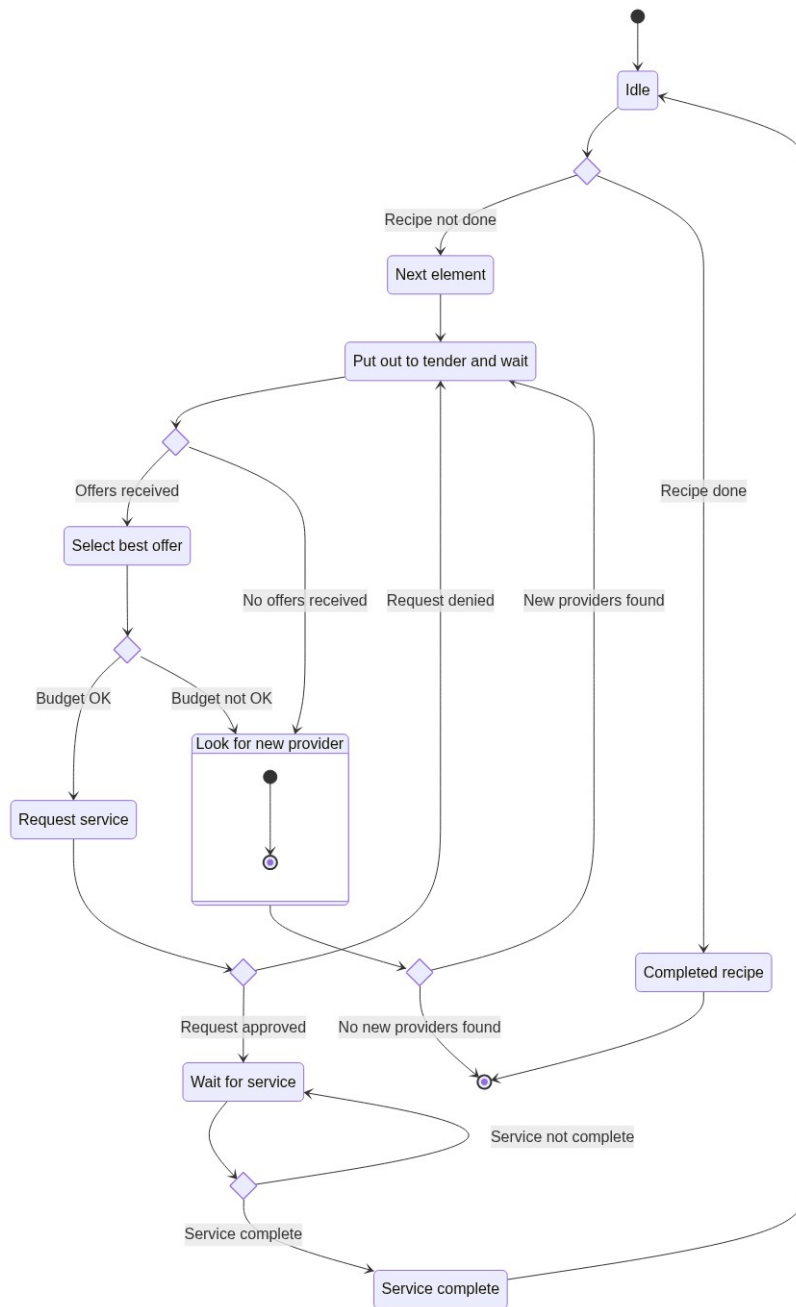


Figure 3.2: Finite-state machine of the service consumer agent

in listing 3.1, with personality profile shown in listing 2.3, has a compatibility score of 0.82 with the *service milestone* gamification technique. Since this particular technique offers a reward that is a trophy, the final compatibility score is 0.85, as shown in listing 3.4.

```

1 {"service_milestone": ServiceMilestone(
2   "service_milestone",
3   pprofile.service_milestone,
4   goal={"metric": "total_services", "target": 2, "comparison": "gte"},
5   reward_item=Item(
6     "Service Trophy", {"effect": "reputation_boost", "value": 10})}}

```

Listing 3.2: Service milestone gamification technique instantiation

```

1 service_milestone = PersonalityProfile(np.array([
2   0.50, 0.45, 0.55, 0.65, 0.50, 0.55,      # Openness
3   0.85, 0.80, 0.75, 0.95, 0.90, 0.80,      # Conscientiousness
4   0.70, 0.65, 0.60, 0.70, 0.65, 0.75,      # Extraversion
5   0.75, 0.70, 0.65, 0.70, 0.60, 0.70,      # Agreeableness
6   0.25, 0.20, 0.20, 0.30, 0.30, 0.25      # Neuroticism
7 ]))

```

Listing 3.3: Service milestone gamification technique personality profile

```

1 INFO:utils.logger: [service milestone] Compatibility: 0.822818360620144
2 INFO:utils.logger: [service milestone] Trophy/badge compatibility for provider2@localhost: 0.91
3 INFO:utils.logger: [service milestone] Combined compatibility: 0.85

```

Listing 3.4: Service milestone gamification technique compatibility calculation

On the other hand, some of the implemented gamification techniques can affect the offer in a more significant way. For example, the *quality focus* gamification technique, which may be considered as an instance of role-playing, increases the duration of the service by a specific amount that depends on the agent's *Dutifulness* facet. Longer service duration in this case denotes better quality. On the other hand, the agent's capacity to provide the service is limited by the number of concurrent services the agent can provide. After calculating the compatibility, agents will ignore this particular gamification technique if the combined compatibility is below a certain threshold.

In the implemented system, we have experimented with various compatibility thresholds and settled on a basic threshold of 0.6. This choice emerged from preliminary experiments demonstrating that thresholds lower than 0.6 led to overly frequent behavioural adjustments, while thresholds above 0.6 failed to significantly influence agent interactions. Therefore, 0.6 provided a balanced approach ensuring meaningful yet not excessive adaptations. What is more, the reward system directly impacts agent behaviour by interacting distinctly with personality facets (particularly *Achievement striving* and *Assertiveness*), modifying decisions beyond basic compatibility. For example, agents with high *Achievement striving* are considerably more responsive to gamification techniques providing trophy-based rewards, explicitly motivating behaviours oriented towards goal completion.

Part IV

Appendices

GIVEn Turtle Serialization

```
1  @prefix :
   ↪ <https://raw.githubusercontent.com/AILab-FOI/MAGO/refs/heads/main/Deliverables/Phase%202/Implementation/GIVEn.rdf#>
   ↪ .
2  @prefix dc: <http://purl.org/dc/elements/1.1/> .
3  @prefix cgo: <http://autosemanticgame.eu/ontologies/cgo#> .
4  @prefix owl: <http://www.w3.org/2002/07/owl#> .
5  @prefix rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#> .
6  @prefix xml: <http://www.w3.org/XML/1998/namespace> .
7  @prefix xsd: <http://www.w3.org/2001/XMLSchema#> .
8  @prefix foaf: <http://xmlns.com/foaf/0.1/> .
9  @prefix rdfs: <http://www.w3.org/2000/01/rdf-schema#> .
10 @prefix vann: <http://purl.org/vocab/vann/> .
11 @prefix terms: <http://purl.org/dc/terms/> .
12 @base
   ↪ <https://raw.githubusercontent.com/AILab-FOI/MAGO/refs/heads/main/Deliverables/Phase%202/Implementation/GIVEn.rdf#>
   ↪ .
13
14 <https://raw.githubusercontent.com/AILab-FOI/MAGO/refs/heads/main/Deliverables/Phase%202/Implementation/GIVEn.rdf#
   ↪ rdf:type owl:Ontology ;
15
16
17
18
19
20
   ↪ owl:imports
   ↪ cgo:
   ↪ ,
   ↪ <http://purl.org/net/VideoGameOntology>
   ↪ ,
   ↪ <https://raw.githubusercontent.com/AILab-FOI/MAGO/refs/heads/main/Previous
   ↪ ;
   ↪ dc:description
   ↪ "This
   ↪ is
   ↪ an
   ↪ ontology
   ↪ that
   ↪ combines
   ↪ video
   ↪ game-related
   ↪ concepts
   ↪ with
   ↪ concepts
   ↪ related
   ↪ to
   ↪ organisational
   ↪ modelling
   ↪ of
   ↪ multi-agent
   ↪ systems
   ↪ as
   ↪ intelligent
   ↪ virtual
   ↪ environments."
   ↪ ;
   ↪ dc:title
   ↪ "Games
   ↪ in
   ↪ Intelligent
   ↪ Virtual
   ↪ Environments
   ↪ Ontology
   ↪ (GIVEn)"
   ↪ ;
   ↪ terms:description
   ↪ "This
   ↪ is
   ↪ an
   ↪ ontology
   ↪ that
   ↪ combines
   ↪ video
   ↪ game-related
   ↪ concepts
   ↪ with
   ↪ concepts
   ↪ related
   ↪ to
   ↪ organisational
   ↪ modelling
   ↪ of
   ↪ multi-agent
   ↪ systems
   ↪ as
   ↪ intelligent
   ↪ virtual
   ↪ environments."
   ↪ ;
```

```

21
    ↪ terms:partOf
    ↪ "This
    ↪ study
    ↪ is
    ↪ supported
    ↪ by
    ↪ the
    ↪ project
    ↪ MOB0DL-2023-08-5618
    ↪ funded
    ↪ by
    ↪ the
    ↪ European
    ↪ Union
    ↪ and
    ↪ the
    ↪ Croatian
    ↪ Science
    ↪ Foundation."
    ↪ ;

22
    ↪ terms:title
    ↪ "Games
    ↪ in
    ↪ Intelligent
    ↪ Virtual
    ↪ Environments
    ↪ Ontology
    ↪ (GIVEn)"
    ↪ ;

23
    ↪ rdfs:comment
    ↪ "This
    ↪ is
    ↪ an
    ↪ ontology
    ↪ that
    ↪ combines
    ↪ video
    ↪ game-related
    ↪ concepts
    ↪ with
    ↪ concepts
    ↪ related
    ↪ to
    ↪ organisational
    ↪ modelling
    ↪ of
    ↪ multi-agent
    ↪ systems
    ↪ as
    ↪ intelligent
    ↪ virtual
    ↪ environments."
    ↪ ;

24
    ↪ foaf:homepage
    ↪ "https://github.com/AILab-FOI/MAGO"
    ↪ ;

25
    ↪ foaf:name
    ↪ "Games
    ↪ in
    ↪ Intelligent
    ↪ Virtual
    ↪ Environments
    ↪ Ontology
    ↪ (GIVEn)"
    ↪ .

26
27 #####
28 # Object Properties
29 #####
30
31 ###
32 ↪ https://raw.githubusercontent.com/AILab-FOI/MAGO/refs/heads/main/Deliverables/Phase%202/Implementation/GIVEn.rdf#OWLObjectProperty_0658272e_b138_4c80_96d5_1666a2a8e370
33 :OWLObjectProperty_0658272e_b138_4c80_96d5_1666a2a8e370 rdfs:type owl:ObjectProperty ;
34 rdfs:domain :OWLClass_1dcf5fa9_2d15_4876_8912_5a3079ab8ea3 ;
    ↪ rdfs:range
    ↪ <http://personales.upv.es/ccarrasc/ooooaflsmas#Objective>
    ↪ ;
    ↪ rdfs:label "consists of objectives"@en-gb .

35
36
37 ###
38 ↪ https://raw.githubusercontent.com/AILab-FOI/MAGO/refs/heads/main/Deliverables/Phase%202/Implementation/GIVEn.rdf#OWLObjectProperty_0fb074df_59e7_46e3_873f_42ac624a80cc
39 :OWLObjectProperty_0fb074df_59e7_46e3_873f_42ac624a80cc rdfs:type owl:ObjectProperty ;
40 rdfs:domain
    ↪ <http://personales.upv.es/ccarrasc/JaCalIVE_Ontology#Observable_Event>
    ↪ ;
    ↪ rdfs:range <http://purl.org/net/VideoGameOntology#Achievement>
    ↪ ;
    ↪ rdfs:label "provides achievement"@en-gb .

41
42
43
44
45 ###
46 ↪ https://raw.githubusercontent.com/AILab-FOI/MAGO/refs/heads/main/Deliverables/Phase%202/Implementation/GIVEn.rdf#OWLObjectProperty_1b06a07c_e2fe_4d5c_affb_36d050530a9c
47 :OWLObjectProperty_1b06a07c_e2fe_4d5c_affb_36d050530a9c rdfs:type owl:ObjectProperty ;
48 rdfs:domain cgo:Narrative ;
    ↪ rdfs:range
    ↪ <http://personales.upv.es/ccarrasc/JaCalIVE_Ontology#Workspace>
    ↪ ;
    ↪ rdfs:label "depends on workspace"@en-gb .

49
50
51
52 ###
53 ↪ https://raw.githubusercontent.com/AILab-FOI/MAGO/refs/heads/main/Deliverables/Phase%202/Implementation/GIVEn.rdf#OWLObjectProperty_1c340508_ab6d_4a77_b307_913a61c0f36d
54 :OWLObjectProperty_1c340508_ab6d_4a77_b307_913a61c0f36d rdfs:type owl:ObjectProperty ;
    ↪ rdfs:domain
    ↪ <http://personales.upv.es/ccarrasc/JaCalIVE_Ontology#Agent>
    ↪ ;
    ↪ rdfs:range :OWLClass_0c864038_db01_4f7d_87c0_650ad7f84af1 ;
    ↪ rdfs:label "is described using"@en-gb .

55
56
57

```

```

58
59 ###
60 ↪ https://raw.githubusercontent.com/AILab-FOI/MAG0/refs/heads/main/Deliverables/Phase%202/Implementation/GIVEn.rdf#OWLObjectProperty_5230412c_73be_4291_839e_734a671a5b28
61 :OWLObjectProperty_5230412c_73be_4291_839e_734a671a5b28 rdfs:type owl:ObjectProperty ;
62 rdfs:domain
63 ↪ <http://personales.upv.es/ccarrasc/JaCalIVE_Ontology#Observable_Event>
64 ↪ ;
65 rdfs:range :OWLClass_1dcf5fa9_2d15_4876_8912_5a3079ab8ea3 ;
66 rdfs:label "depends on"@en-gb .
67
68 ###
69 ↪ https://raw.githubusercontent.com/AILab-FOI/MAG0/refs/heads/main/Deliverables/Phase%202/Implementation/GIVEn.rdf#OWLObjectProperty_5daa59b7_9bbb_40b5_800a_2ab2f3537424
70 :OWLObjectProperty_5daa59b7_9bbb_40b5_800a_2ab2f3537424 rdfs:type owl:ObjectProperty ;
71 rdfs:domain
72 ↪ <http://purl.org/net/VideoGameOntology#Achievement> ;
73 rdfs:range :OWLClass_0c864038_db01_4f7d_87c0_650ad7f84af1 ;
74 rdfs:label "affects inventory"@en-gb .
75
76 ###
77 ↪ https://raw.githubusercontent.com/AILab-FOI/MAG0/refs/heads/main/Deliverables/Phase%202/Implementation/GIVEn.rdf#OWLObjectProperty_88097da3_4059_4b9c_8161_89d2417f8a5e
78 :OWLObjectProperty_88097da3_4059_4b9c_8161_89d2417f8a5e rdfs:type owl:ObjectProperty ;
79 rdfs:domain
80 ↪ <http://personales.upv.es/ccarrasc/JaCalIVE_Ontology#Observable_Event>
81 ↪ ;
82 rdfs:range cgo:Narrative ;
83 rdfs:label "affects narrative"@en-gb .
84
85 ###
86 ↪ https://raw.githubusercontent.com/AILab-FOI/MAG0/refs/heads/main/Deliverables/Phase%202/Implementation/GIVEn.rdf#OWLObjectProperty_eb2068d8_392c_4fd7_9ddf_31245a7f9a4d
87 :OWLObjectProperty_eb2068d8_392c_4fd7_9ddf_31245a7f9a4d rdfs:type owl:ObjectProperty ;
88 rdfs:domain
89 ↪ <http://personales.upv.es/ccarrasc/JaCalIVE_Ontology#Agent>
90 ↪ ;
91 rdfs:range
92 ↪ <http://personales.upv.es/ccarrasc/JaCalIVE_Ontology#Observable_Property>
93 ↪ ;
94 rdfs:label "has observable property"@en-gb .
95
96 #####
97 # Classes
98 #####
99
100 ### http://autosemanticgame.eu/ontologies/cgo#Audio
101 cgo:Audio rdfs:subClassOf <http://personales.upv.es/ccarrasc/JaCalIVE_Ontology#IVE_Artifact> .
102
103 ### http://autosemanticgame.eu/ontologies/cgo#Game
104 cgo:Game owl:equivalentClass <http://purl.org/net/VideoGameOntology#Game> ;
105 rdfs:subClassOf <http://personales.upv.es/ccarrasc/JaCalIVE_Ontology#IVE> .
106
107 ### http://autosemanticgame.eu/ontologies/cgo#GameDesign
108 cgo:GameDesign rdfs:subClassOf <http://personales.upv.es/ccarrasc/ooooaflsmas#Norm> .
109
110 ### http://autosemanticgame.eu/ontologies/cgo#Gameplay
111 cgo:Gameplay rdfs:subClassOf <http://personales.upv.es/ccarrasc/ooooaflsmas#Strategy> .
112
113 ### http://autosemanticgame.eu/ontologies/cgo#LevelDesign
114 cgo:LevelDesign rdfs:subClassOf <http://personales.upv.es/ccarrasc/JaCalIVE_Ontology#IVE_Artifact> .
115
116 ### http://autosemanticgame.eu/ontologies/cgo#Visuals
117 cgo:Visuals rdfs:subClassOf <http://personales.upv.es/ccarrasc/JaCalIVE_Ontology#IVE_Artifact> .
118
119 ### http://personales.upv.es/ccarrasc/ooooaflsmas#Objective
120 <http://personales.upv.es/ccarrasc/ooooaflsmas#Objective> rdfs:subClassOf
121 ↪ :OWLClass_1dcf5fa9_2d15_4876_8912_5a3079ab8ea3 .
122
123 ### http://purl.org/net/VideoGameOntology#Achievement
124 <http://purl.org/net/VideoGameOntology#Achievement> rdfs:subClassOf
125 ↪ <http://personales.upv.es/ccarrasc/JaCalIVE_Ontology#Artifact> .
126
127 ### http://purl.org/net/VideoGameOntology#Character
128 <http://purl.org/net/VideoGameOntology#Character> rdfs:subClassOf
129 ↪ <http://personales.upv.es/ccarrasc/JaCalIVE_Ontology#Agent> .
130
131 ### http://purl.org/net/VideoGameOntology#Feature
132 <http://purl.org/net/VideoGameOntology#Feature> rdfs:subClassOf
133 ↪ <http://personales.upv.es/ccarrasc/JaCalIVE_Ontology#Observable_Property> .
134
135 ### http://purl.org/net/VideoGameOntology#Game
136 <http://purl.org/net/VideoGameOntology#Game> rdfs:subClassOf <http://personales.upv.es/ccarrasc/JaCalIVE_Ontology#IVE>
137 ↪ .
138
139 ### http://purl.org/net/VideoGameOntology#InstantaneousEvent
140 <http://purl.org/net/VideoGameOntology#InstantaneousEvent> rdfs:subClassOf
141 ↪ <http://personales.upv.es/ccarrasc/JaCalIVE_Ontology#Observable_Event> .
142
143 ### http://purl.org/net/VideoGameOntology#Item
144 <http://purl.org/net/VideoGameOntology#Item> rdfs:subClassOf
145 ↪ <http://personales.upv.es/ccarrasc/JaCalIVE_Ontology#Physical_Artifact> .
146
147 ### http://purl.org/net/VideoGameOntology#Player
148 <http://purl.org/net/VideoGameOntology#Player> rdfs:subClassOf
149 ↪ <http://personales.upv.es/ccarrasc/JaCalIVE_Ontology#Human_Immersed_Agent> .
150
151 ### http://purl.org/net/VideoGameOntology#PlayingArea
152 <http://purl.org/net/VideoGameOntology#PlayingArea> rdfs:subClassOf
153 ↪ <http://personales.upv.es/ccarrasc/JaCalIVE_Ontology#IVE_Workspace> .

```

```

152   ###
153   ↪ https://raw.githubusercontent.com/AILab-FOI/MAGO/refs/heads/main/Deliverables/Phase%202/Implementation/GIVEn.rdf#OWLClass_0c864038_db01_4f7d_87c0_650ad7f84af1
154   :OWLClass_0c864038_db01_4f7d_87c0_650ad7f84af1 rdf:type owl:Class ;
155                                           rdfs:subClassOf
156                                           ↪ <http://personales.upv.es/ccarrasc/ooooafslmas#KnowledgeArtifact> ;
157                                           rdfs:label "Inventory"@en-gb .
158
159   ###
160   ↪ https://raw.githubusercontent.com/AILab-FOI/MAGO/refs/heads/main/Deliverables/Phase%202/Implementation/GIVEn.rdf#OWLClass_1dcf5fa9_2d15_4876_8912_5a3079ab8ea3
161   :OWLClass_1dcf5fa9_2d15_4876_8912_5a3079ab8ea3 rdf:type owl:Class ;
162                                           rdfs:label "Quest"@en-gb .
163
164   ###
165   ↪ https://raw.githubusercontent.com/AILab-FOI/MAGO/refs/heads/main/Deliverables/Phase%202/Implementation/GIVEn.rdf#OWLClass_455b23c2_5f8a_4397_841f_33132d3b92a2
166   :OWLClass_455b23c2_5f8a_4397_841f_33132d3b92a2 rdf:type owl:Class ;
167                                           rdfs:subClassOf
168                                           ↪ <http://personales.upv.es/ccarrasc/JaCalIVE_Ontology#IVE_Law> ,
169                                           :OWLClass_b16d01eb_7c6a_47a7_b7a4_32d3f83f30f3 ;
170                                           rdfs:label "Local rule"@en-gb .
171
172   ###
173   ↪ https://raw.githubusercontent.com/AILab-FOI/MAGO/refs/heads/main/Deliverables/Phase%202/Implementation/GIVEn.rdf#OWLClass_b16d01eb_7c6a_47a7_b7a4_32d3f83f30f3
174   :OWLClass_b16d01eb_7c6a_47a7_b7a4_32d3f83f30f3 rdf:type owl:Class ;
175                                           rdfs:subClassOf <http://personales.upv.es/ccarrasc/ooooafslmas#Norm> ;
176                                           rdfs:label "Rule"@en-gb .
177
178   ### Generated by the OWL API (version 4.5.29.2024-05-13T12:11:03Z) https://github.com/owlcs/owlapi

```

Listing 3.5: Turtle serialization of the GIVEn ontology

GIVEn OWL/XML Serialization

```
1 <?xml version="1.0"?>
2 <Ontology xmlns="http://www.w3.org/2002/07/owl#"
3
4   ↳ xml:base="https://raw.githubusercontent.com/AILab-FOI/MAGO/refs/heads/main/Deliverables/Phase%202/Implementation/GIVEn.rdf"
5   xmlns:rdf="http://www.w3.org/1999/02/22-rdf-syntax-ns#"
6   xmlns:xml="http://www.w3.org/XML/1998/namespace"
7   xmlns:xsd="http://www.w3.org/2001/XMLSchema#"
8   xmlns:rdfs="http://www.w3.org/2000/01/rdf-schema#"
9
10  ↳ ontologyIRI="https://raw.githubusercontent.com/AILab-FOI/MAGO/refs/heads/main/Deliverables/Phase%202/Implementation/GIVEn.rdf#"
11  <Prefix name=""
12  ↳ IRI="https://raw.githubusercontent.com/AILab-FOI/MAGO/refs/heads/main/Deliverables/Phase%202/Implementation/GIVEn.rdf#" />
13  <Prefix name="dc" IRI="http://purl.org/dc/elements/1.1/" />
14  <Prefix name="cgo" IRI="http://autosemanticgame.eu/ontologies/cgo#" />
15  <Prefix name="owl" IRI="http://www.w3.org/2002/07/owl#" />
16  <Prefix name="rdf" IRI="http://www.w3.org/1999/02/22-rdf-syntax-ns#" />
17  <Prefix name="xml" IRI="http://www.w3.org/XML/1998/namespace#" />
18  <Prefix name="xsd" IRI="http://www.w3.org/2001/XMLSchema#" />
19  <Prefix name="foaf" IRI="http://xmlns.com/foaf/0.1/" />
20  <Prefix name="rdfs" IRI="http://www.w3.org/2000/01/rdf-schema#" />
21  <Prefix name="vann" IRI="http://purl.org/vocab/vann/" />
22  <Prefix name="terms" IRI="http://purl.org/dc/terms/" />
23  <Import>http://autosemanticgame.eu/ontologies/cgo#</Import>
24  <Import>http://purl.org/net/VideoGameOntology</Import>
25  <Import>https://raw.githubusercontent.com/AILab-FOI/MAGO/refs/heads/main/Previous%20Work/MAMB05.owl</Import>
26  <Annotation>
27    <AnnotationProperty abbreviatedIRI="dc:description"/>
28    <Literal>This is an ontology that combines video game-related concepts with concepts related to organisational
29    ↳ modelling of multi-agent systems as intelligent virtual environments.</Literal>
30  </Annotation>
31  <Annotation>
32    <AnnotationProperty abbreviatedIRI="dc:title"/>
33    <Literal>Games in Intelligent Virtual Environments Ontology (GIVEn)</Literal>
34  </Annotation>
35  <Annotation>
36    <AnnotationProperty abbreviatedIRI="terms:description"/>
37    <Literal>This is an ontology that combines video game-related concepts with concepts related to organisational
38    ↳ modelling of multi-agent systems as intelligent virtual environments.</Literal>
39  </Annotation>
40  <Annotation>
41    <AnnotationProperty abbreviatedIRI="terms:partOf"/>
42    <Literal>This study is supported by the project MOB00L-2023-08-5618 funded by the European Union and the
43    ↳ Croatian Science Foundation.</Literal>
44  </Annotation>
45  <Annotation>
46    <AnnotationProperty abbreviatedIRI="terms:title"/>
47    <Literal>Games in Intelligent Virtual Environments Ontology (GIVEn)</Literal>
48  </Annotation>
49  <Annotation>
50    <AnnotationProperty abbreviatedIRI="rdfs:comment"/>
51    <Literal>This is an ontology that combines video game-related concepts with concepts related to organisational
52    ↳ modelling of multi-agent systems as intelligent virtual environments.</Literal>
53  </Annotation>
54  <Annotation>
55    <AnnotationProperty abbreviatedIRI="foaf:homepage"/>
56    <Literal>https://github.com/AILab-FOI/MAGO</Literal>
57  </Annotation>
58  <Annotation>
59    <AnnotationProperty abbreviatedIRI="foaf:name"/>
60    <Literal>Games in Intelligent Virtual Environments Ontology (GIVEn)</Literal>
61  </Annotation>
62  <Declaration>
63    <Class IRI="#OWLClass_0c864038_db01_4f7d_87c0_650ad7f84af1"/>
64  </Declaration>
65  <Declaration>
66    <Class IRI="#OWLClass_1dcf5fa9_2d15_4876_8912_5a3079ab8ea3"/>
67  </Declaration>
68  <Declaration>
69    <Class IRI="#OWLClass_455b23c2_5f8a_4397_841f_33132d3b92a2"/>
70  </Declaration>
71  <Declaration>
72    <Class IRI="#OWLClass_b16d01eb_7c6a_47a7_b7a4_32d3f83f30f3"/>
73  </Declaration>
74  <Declaration>
75    <ObjectProperty IRI="#OWLObjectProperty_0658272e_b138_4c80_96d5_1666a2a8e370"/>
76  </Declaration>
77  <Declaration>
78    <ObjectProperty IRI="#OWLObjectProperty_0fb074df_59e7_46e3_873f_42ac624a80cc"/>
79  </Declaration>
80  <Declaration>
81    <ObjectProperty IRI="#OWLObjectProperty_1b06a07c_e2fe_4d5c_affb_36d050530a9c"/>
82  </Declaration>
83  <Declaration>
84    <ObjectProperty IRI="#OWLObjectProperty_1c340508_ab6d_4a77_b307_913a61c0f36d"/>
85  </Declaration>
86  <Declaration>
87    <ObjectProperty IRI="#OWLObjectProperty_5230412c_73be_4291_839e_734a671a5b28"/>
88  </Declaration>
89  <Declaration>
90    <ObjectProperty IRI="#OWLObjectProperty_5daa59b7_9bbb_40b5_800a_2ab2f3537424"/>
91  </Declaration>
92  <Declaration>
93    <ObjectProperty IRI="#OWLObjectProperty_88097da3_4059_4b9c_8161_89d2417f8a5e"/>
94  </Declaration>
```

```

88 <Declaration>
89   <ObjectProperty IRI="#OWLObjectProperty_eb2068d8_392c_4fd7_9ddf_31245a7f9a4d"/>
90 </Declaration>
91 <Declaration>
92   <AnnotationProperty abbreviatedIRI="terms:license"/>
93 </Declaration>
94 <Declaration>
95   <AnnotationProperty abbreviatedIRI="dc:title"/>
96 </Declaration>
97 <EquivalentClasses>
98   <Class abbreviatedIRI="cgo:Game"/>
99   <Class IRI="http://purl.org/net/VideoGameOntology#Game"/>
100 </EquivalentClasses>
101 <SubClassOf>
102   <Class abbreviatedIRI="cgo:Audio"/>
103   <Class IRI="http://personales.upv.es/ccarrasc/JaCalIVE_Ontology#IVE_Artifact"/>
104 </SubClassOf>
105 <SubClassOf>
106   <Class abbreviatedIRI="cgo:Game"/>
107   <Class IRI="http://personales.upv.es/ccarrasc/JaCalIVE_Ontology#IVE"/>
108 </SubClassOf>
109 <SubClassOf>
110   <Class abbreviatedIRI="cgo:GameDesign"/>
111   <Class IRI="http://personales.upv.es/ccarrasc/ooooflsmas#Norm"/>
112 </SubClassOf>
113 <SubClassOf>
114   <Class abbreviatedIRI="cgo:Gameplay"/>
115   <Class IRI="http://personales.upv.es/ccarrasc/ooooflsmas#Strategy"/>
116 </SubClassOf>
117 <SubClassOf>
118   <Class abbreviatedIRI="cgo:LevelDesign"/>
119   <Class IRI="http://personales.upv.es/ccarrasc/JaCalIVE_Ontology#IVE_Artifact"/>
120 </SubClassOf>
121 <SubClassOf>
122   <Class abbreviatedIRI="cgo:Visuals"/>
123   <Class IRI="http://personales.upv.es/ccarrasc/JaCalIVE_Ontology#IVE_Artifact"/>
124 </SubClassOf>
125 <SubClassOf>
126   <Class IRI="http://personales.upv.es/ccarrasc/ooooflsmas#Objective"/>
127   <Class IRI="#OWLClass_1dcf5fa9_2d15_4876_8912_5a3079ab8ea3"/>
128 </SubClassOf>
129 <SubClassOf>
130   <Class IRI="http://purl.org/net/VideoGameOntology#Achievement"/>
131   <Class IRI="http://personales.upv.es/ccarrasc/JaCalIVE_Ontology#Artifact"/>
132 </SubClassOf>
133 <SubClassOf>
134   <Class IRI="http://purl.org/net/VideoGameOntology#Character"/>
135   <Class IRI="http://personales.upv.es/ccarrasc/JaCalIVE_Ontology#Agent"/>
136 </SubClassOf>
137 <SubClassOf>
138   <Class IRI="http://purl.org/net/VideoGameOntology#Feature"/>
139   <Class IRI="http://personales.upv.es/ccarrasc/JaCalIVE_Ontology#Observable_Property"/>
140 </SubClassOf>
141 <SubClassOf>
142   <Class IRI="http://purl.org/net/VideoGameOntology#Game"/>
143   <Class IRI="http://personales.upv.es/ccarrasc/JaCalIVE_Ontology#IVE"/>
144 </SubClassOf>
145 <SubClassOf>
146   <Class IRI="http://purl.org/net/VideoGameOntology#InstantaneousEvent"/>
147   <Class IRI="http://personales.upv.es/ccarrasc/JaCalIVE_Ontology#Observable_Event"/>
148 </SubClassOf>
149 <SubClassOf>
150   <Class IRI="http://purl.org/net/VideoGameOntology#Item"/>
151   <Class IRI="http://personales.upv.es/ccarrasc/JaCalIVE_Ontology#Physical_Artifact"/>
152 </SubClassOf>
153 <SubClassOf>
154   <Class IRI="http://purl.org/net/VideoGameOntology#Player"/>
155   <Class IRI="http://personales.upv.es/ccarrasc/JaCalIVE_Ontology#Human_Immersed_Agent"/>
156 </SubClassOf>
157 <SubClassOf>
158   <Class IRI="http://purl.org/net/VideoGameOntology#PlayingArea"/>
159   <Class IRI="http://personales.upv.es/ccarrasc/JaCalIVE_Ontology#IVE_Workspace"/>
160 </SubClassOf>
161 <SubClassOf>
162   <Class IRI="#OWLClass_0c864038_db01_4f7d_87c0_650ad7f84af1"/>
163   <Class IRI="http://personales.upv.es/ccarrasc/ooooflsmas#KnowledgeArtifact"/>
164 </SubClassOf>
165 <SubClassOf>
166   <Class IRI="#OWLClass_455b23c2_5f8a_4397_841f_33132d3b92a2"/>
167   <Class IRI="http://personales.upv.es/ccarrasc/JaCalIVE_Ontology#IVE_Law"/>
168 </SubClassOf>
169 <SubClassOf>
170   <Class IRI="#OWLClass_455b23c2_5f8a_4397_841f_33132d3b92a2"/>
171   <Class IRI="#OWLClass_b16d01eb_7c6a_47a7_b7a4_32d3f83f30f3"/>
172 </SubClassOf>
173 <SubClassOf>
174   <Class IRI="#OWLClass_b16d01eb_7c6a_47a7_b7a4_32d3f83f30f3"/>
175   <Class IRI="http://personales.upv.es/ccarrasc/ooooflsmas#Norm"/>
176 </SubClassOf>
177 <ObjectPropertyDomain>
178   <ObjectProperty IRI="#OWLObjectProperty_0658272e_b138_4c80_96d5_1666a2a8e370"/>
179   <Class IRI="#OWLClass_1dcf5fa9_2d15_4876_8912_5a3079ab8ea3"/>
180 </ObjectPropertyDomain>
181 <ObjectPropertyDomain>
182   <ObjectProperty IRI="#OWLObjectProperty_0fb074df_59e7_46e3_873f_42ac624a80cc"/>
183   <Class IRI="http://personales.upv.es/ccarrasc/JaCalIVE_Ontology#Observable_Event"/>
184 </ObjectPropertyDomain>
185 <ObjectPropertyDomain>
186   <ObjectProperty IRI="#OWLObjectProperty_1b06a07c_e2fe_4d5c_affb_36d050530a9c"/>
187   <Class abbreviatedIRI="cgo:Narrative"/>
188 </ObjectPropertyDomain>
189 <ObjectPropertyDomain>
190   <ObjectProperty IRI="#OWLObjectProperty_1c340508_ab6d_4a77_b307_913a61c0f36d"/>
191   <Class IRI="http://personales.upv.es/ccarrasc/JaCalIVE_Ontology#Agent"/>
192 </ObjectPropertyDomain>
193 <ObjectPropertyDomain>
194   <ObjectProperty IRI="#OWLObjectProperty_5230412c_73be_4291_839e_734a671a5b28"/>
195   <Class IRI="http://personales.upv.es/ccarrasc/JaCalIVE_Ontology#Observable_Event"/>
196 </ObjectPropertyDomain>
197 <ObjectPropertyDomain>
198   <ObjectProperty IRI="#OWLObjectProperty_5daa59b7_9bbb_40b5_800a_2ab2f3537424"/>
199   <Class IRI="http://purl.org/net/VideoGameOntology#Achievement"/>
200 </ObjectPropertyDomain>
201 <ObjectPropertyDomain>
202   <ObjectProperty IRI="#OWLObjectProperty_88097da3_4059_4b9c_8161_89d2417f8a5e"/>
203   <Class IRI="http://personales.upv.es/ccarrasc/JaCalIVE_Ontology#Observable_Event"/>
204 </ObjectPropertyDomain>
205 <ObjectPropertyDomain>

```

```

206     <ObjectProperty IRI="#OWLObjectProperty_eb2068d8_392c_4fd7_9ddf_31245a7f9a4d"/>
207     <Class IRI="http://personales.upv.es/ccarrasc/JaCaLIVE.Ontology#Agent"/>
208   </ObjectPropertyDomain>
209   <ObjectPropertyRange>
210     <ObjectProperty IRI="#OWLObjectProperty_0658272e_b138_4c80_96d5_1666a2a8e370"/>
211     <Class IRI="http://personales.upv.es/ccarrasc/ooooaflsmas#Objective"/>
212   </ObjectPropertyRange>
213   <ObjectPropertyRange>
214     <ObjectProperty IRI="#OWLObjectProperty_0fb074df_59e7_46e3_873f_42ac624a80cc"/>
215     <Class IRI="http://purl.org/net/VideoGameOntology#Achievement"/>
216   </ObjectPropertyRange>
217   <ObjectPropertyRange>
218     <ObjectProperty IRI="#OWLObjectProperty_1b06a07c_e2fe_4d5c_affb_36d050530a9c"/>
219     <Class IRI="http://personales.upv.es/ccarrasc/JaCaLIVE.Ontology#Workspace"/>
220   </ObjectPropertyRange>
221   <ObjectPropertyRange>
222     <ObjectProperty IRI="#OWLObjectProperty_1c340508_ab6d_4a77_b307_913a61c0f36d"/>
223     <Class IRI="#OWLClass_0c864038_db01_4f7d_87c0_650ad7f84af1"/>
224   </ObjectPropertyRange>
225   <ObjectPropertyRange>
226     <ObjectProperty IRI="#OWLObjectProperty_5230412c_73be_4291_839e_734a671a5b28"/>
227     <Class IRI="#OWLClass_1dcf5fa9_2d15_4876_8912_5a3079ab8ea3"/>
228   </ObjectPropertyRange>
229   <ObjectPropertyRange>
230     <ObjectProperty IRI="#OWLObjectProperty_5daa59b7_9bbb_40b5_800a_2ab2f3537424"/>
231     <Class IRI="#OWLClass_0c864038_db01_4f7d_87c0_650ad7f84af1"/>
232   </ObjectPropertyRange>
233   <ObjectPropertyRange>
234     <ObjectProperty IRI="#OWLObjectProperty_88097da3_4059_4b9c_8161_89d2417f8a5e"/>
235     <Class abbreviatedIRI="cgo:Narrative"/>
236   </ObjectPropertyRange>
237   <ObjectPropertyRange>
238     <ObjectProperty IRI="#OWLObjectProperty_eb2068d8_392c_4fd7_9ddf_31245a7f9a4d"/>
239     <Class IRI="http://personales.upv.es/ccarrasc/JaCaLIVE.Ontology#Observable_Property"/>
240   </ObjectPropertyRange>
241   <AnnotationAssertion>
242     <AnnotationProperty abbreviatedIRI="rdfs:label"/>
243     <IRI>#OWLClass_0c864038_db01_4f7d_87c0_650ad7f84af1</IRI>
244     <Literal xml:lang="en-gb">Inventory</Literal>
245   </AnnotationAssertion>
246   <AnnotationAssertion>
247     <AnnotationProperty abbreviatedIRI="rdfs:label"/>
248     <IRI>#OWLClass_1dcf5fa9_2d15_4876_8912_5a3079ab8ea3</IRI>
249     <Literal xml:lang="en-gb">Quest</Literal>
250   </AnnotationAssertion>
251   <AnnotationAssertion>
252     <AnnotationProperty abbreviatedIRI="rdfs:label"/>
253     <IRI>#OWLClass_455b23c2_5f8a_4397_841f_33132d3b92a2</IRI>
254     <Literal xml:lang="en-gb">Local_rule</Literal>
255   </AnnotationAssertion>
256   <AnnotationAssertion>
257     <AnnotationProperty abbreviatedIRI="rdfs:label"/>
258     <IRI>#OWLClass_b16d01eb_7c6a_47a7_b7a4_32d3f83f30f3</IRI>
259     <Literal xml:lang="en-gb">Rule</Literal>
260   </AnnotationAssertion>
261   <AnnotationAssertion>
262     <AnnotationProperty abbreviatedIRI="rdfs:label"/>
263     <IRI>#OWLObjectProperty_0658272e_b138_4c80_96d5_1666a2a8e370</IRI>
264     <Literal xml:lang="en-gb">consists of objectives</Literal>
265   </AnnotationAssertion>
266   <AnnotationAssertion>
267     <AnnotationProperty abbreviatedIRI="rdfs:label"/>
268     <IRI>#OWLObjectProperty_0fb074df_59e7_46e3_873f_42ac624a80cc</IRI>
269     <Literal xml:lang="en-gb">provides achievement</Literal>
270   </AnnotationAssertion>
271   <AnnotationAssertion>
272     <AnnotationProperty abbreviatedIRI="rdfs:label"/>
273     <IRI>#OWLObjectProperty_1b06a07c_e2fe_4d5c_affb_36d050530a9c</IRI>
274     <Literal xml:lang="en-gb">depends on workspace</Literal>
275   </AnnotationAssertion>
276   <AnnotationAssertion>
277     <AnnotationProperty abbreviatedIRI="rdfs:label"/>
278     <IRI>#OWLObjectProperty_1c340508_ab6d_4a77_b307_913a61c0f36d</IRI>
279     <Literal xml:lang="en-gb">is described using</Literal>
280   </AnnotationAssertion>
281   <AnnotationAssertion>
282     <AnnotationProperty abbreviatedIRI="rdfs:label"/>
283     <IRI>#OWLObjectProperty_5230412c_73be_4291_839e_734a671a5b28</IRI>
284     <Literal xml:lang="en-gb">depends on</Literal>
285   </AnnotationAssertion>
286   <AnnotationAssertion>
287     <AnnotationProperty abbreviatedIRI="rdfs:label"/>
288     <IRI>#OWLObjectProperty_5daa59b7_9bbb_40b5_800a_2ab2f3537424</IRI>
289     <Literal xml:lang="en-gb">affects inventory</Literal>
290   </AnnotationAssertion>
291   <AnnotationAssertion>
292     <AnnotationProperty abbreviatedIRI="rdfs:label"/>
293     <IRI>#OWLObjectProperty_88097da3_4059_4b9c_8161_89d2417f8a5e</IRI>
294     <Literal xml:lang="en-gb">affects narrative</Literal>
295   </AnnotationAssertion>
296   <AnnotationAssertion>
297     <AnnotationProperty abbreviatedIRI="rdfs:label"/>
298     <IRI>#OWLObjectProperty_eb2068d8_392c_4fd7_9ddf_31245a7f9a4d</IRI>
299     <Literal xml:lang="en-gb">has observable property</Literal>
300   </AnnotationAssertion>
301 </Ontology>
302
303
304
305 <!-- Generated by the OWL API (version 4.5.29.2024-05-13T12:11:03Z) https://github.com/owlcs/owlapi -->

```

Listing 3.6: OWL/XML serialization of the GIVEN ontology

MAGO-Ga Turtle Serialization

```
1 @prefix : <http://www.semanticweb.org/bogdan/ontologies/2025/5/untitled-ontology-91/> .
2 @prefix owl: <http://www.w3.org/2002/07/owl#> .
3 @prefix rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#> .
4 @prefix xml: <http://www.w3.org/XML/1998/namespace> .
5 @prefix xsd: <http://www.w3.org/2001/XMLSchema#> .
6 @prefix rdfs: <http://www.w3.org/2000/01/rdf-schema#> .
7 @base <http://www.semanticweb.org/bogdan/ontologies/2025/5/untitled-ontology-91/> .
8
9 <https://raw.githubusercontent.com/AILab-FOI/MAGO/refs/heads/main/Deliverables/Phase%202/Implementation/MAGO-Ga.owx>
10 ↳ rdf:type owl:Ontology ;
11
12
13
14 #####
15 # Object Properties
16 #####
17
18 ###
19 ↳ https://raw.githubusercontent.com/AILab-FOI/MAGO/refs/heads/main/Deliverables/Phase%202/Implementation/MAGO-Ga.owx#OWLObjectProperty_033c65dc_4b0b_49a0_9bde_461467eb76d6
20 :OWLObjectProperty_033c65dc_4b0b_49a0_9bde_461467eb76d6 rdf:type owl:ObjectProperty ;
21 rdfs:domain :OWLClass_d7900a7c_c25b_4ea3_ad4e_8716bbf24045 ;
22 rdfs:range <http://purl.org/net/VideoGameOntology#GainEvent> ;
23 rdfs:label "has gamification outcome"@en-gb .
24
25 ###
26 ↳ https://raw.githubusercontent.com/AILab-FOI/MAGO/refs/heads/main/Deliverables/Phase%202/Implementation/MAGO-Ga.owx#OWLObjectProperty_268d7ca3_6f86_4e57_9627_e0320f1d33d1
27 :OWLObjectProperty_268d7ca3_6f86_4e57_9627_e0320f1d33d1 rdf:type owl:ObjectProperty ;
28 rdfs:domain :OWLClass_d7900a7c_c25b_4ea3_ad4e_8716bbf24045 ;
29 rdfs:range :OWLClass_6dae9e4d_d430_445c_b054_d2db632a0da9 ;
30 rdfs:label "has gamification goal"@en-gb .
31
32 ###
33 ↳ https://raw.githubusercontent.com/AILab-FOI/MAGO/refs/heads/main/Deliverables/Phase%202/Implementation/MAGO-Ga.owx#OWLObjectProperty_37fa3a36_2699_4e83_b0d7_8b8e6349fd73
34 :OWLObjectProperty_37fa3a36_2699_4e83_b0d7_8b8e6349fd73 rdf:type owl:ObjectProperty ,
35 owl:FunctionalProperty ;
36 rdfs:domain :OWLClass_0613fb28_22ee_4c72_939f_4344f65624d7 ;
37 rdfs:range :OWLClass_b3f8233f_1eca_4632_895f_8ec633cf37e4 ;
38 rdfs:label "has personality feature"@en-gb .
39
40 ###
41 ↳ https://raw.githubusercontent.com/AILab-FOI/MAGO/refs/heads/main/Deliverables/Phase%202/Implementation/MAGO-Ga.owx#OWLObjectProperty_67cb4fc7_8aa5_4668_a353_308e7d7b6dd1
42 :OWLObjectProperty_67cb4fc7_8aa5_4668_a353_308e7d7b6dd1 rdf:type owl:ObjectProperty ;
43 owl:inverseOf
44 ↳ :OWLObjectProperty_68e185c9_f82f_4507_8f25_4d50e8b3cc3c ;
45 rdfs:label "defines feature"@en-gb .
46
47 ###
48 ↳ https://raw.githubusercontent.com/AILab-FOI/MAGO/refs/heads/main/Deliverables/Phase%202/Implementation/MAGO-Ga.owx#OWLObjectProperty_68e185c9_f82f_4507_8f25_4d50e8b3cc3c
49 :OWLObjectProperty_68e185c9_f82f_4507_8f25_4d50e8b3cc3c rdf:type owl:ObjectProperty ;
50 rdfs:domain :OWLClass_b3f8233f_1eca_4632_895f_8ec633cf37e4 ;
51 rdfs:range :OWLClass_161fdab5_a39e_42b4_9cee_84a03f52ee88 ;
52 rdfs:label "is a part of model"@en-gb .
53
54 ###
55 ↳ https://raw.githubusercontent.com/AILab-FOI/MAGO/refs/heads/main/Deliverables/Phase%202/Implementation/MAGO-Ga.owx#OWLObjectProperty_8c046b10_bc4c_4cb2_aaf8_98113094df12
56 :OWLObjectProperty_8c046b10_bc4c_4cb2_aaf8_98113094df12 rdf:type owl:ObjectProperty ;
```

```

55         rdfs:domain
56         ↪ <http://personales.upv.es/ccarrasc/JaCalIVE_Ontology#Observable_Event>
57         ↪ ;
58         rdfs:range
59         ↪ <http://personales.upv.es/ccarrasc/JaCalIVE_Ontology#Physical_Artifact>
60         ↪ ;
61         rdfs:label "has event object"@en-gb .
62
63 #####
64 ↪ https://raw.githubusercontent.com/AILab-FOI/MAGO/refs/heads/main/Deliverables/Phase%202/Implementation/MAGO-Ga.owx#OWLObjectProperty_a5d640f9_c2fc_479c_ab6a_0d354f65e652
65 :OWLObjectProperty_a5d640f9_c2fc_479c_ab6a_0d354f65e652 rdfs:type owl:ObjectProperty ;
66         owl:FunctionalProperty ;
67         rdfs:domain [ rdfs:type owl:Class ;
68         owl:unionOf (
69         ↪ <http://personales.upv.es/ccarrasc/JaCalIVE_Ontology#Agent>
70         ↪ :OWLClass_0613fb28_22ee_4c72_939f_4344f65624d7
71         ↪ :OWLClass_d7900a7c_c25b_4ea3_ad4e_8716bbf24045
72         )
73         ] ;
74         rdfs:range :OWLClass_19a73670_711c_40c7_a371_230c83aaa5b5 ;
75         rdfs:label "has personality profile"@en-gb .
76
77 #####
78 ↪ https://raw.githubusercontent.com/AILab-FOI/MAGO/refs/heads/main/Deliverables/Phase%202/Implementation/MAGO-Ga.owx#OWLObjectProperty_dcd17b13_165e_4453_a7f5_24ded1c0bc7e
79 :OWLObjectProperty_dcd17b13_165e_4453_a7f5_24ded1c0bc7e rdfs:type owl:ObjectProperty ;
80         rdfs:domain
81         ↪ <http://personales.upv.es/ccarrasc/JaCalIVE_Ontology#Agent>
82         ↪ ;
83         rdfs:range
84         ↪ <http://personales.upv.es/ccarrasc/JaCalIVE_Ontology#Observable_Property>
85         ↪ ;
86         rdfs:label "has observable property"@en-gb .
87
88 #####
89 # Data properties
90 #####
91 #####
92 #####
93 ↪ https://raw.githubusercontent.com/AILab-FOI/MAGO/refs/heads/main/Deliverables/Phase%202/Implementation/MAGO-Ga.owx#OWLDataProperty_40390f82_26b7_4600_a674_c779002bf747
94 :OWLDataProperty_40390f82_26b7_4600_a674_c779002bf747 rdfs:type owl:DatatypeProperty ;
95         owl:FunctionalProperty ;
96         rdfs:domain :OWLClass_0613fb28_22ee_4c72_939f_4344f65624d7 ;
97         rdfs:range xsd:float ;
98         rdfs:label "has weight value"@en-gb .
99
100 #####
101 ↪ https://raw.githubusercontent.com/AILab-FOI/MAGO/refs/heads/main/Deliverables/Phase%202/Implementation/MAGO-Ga.owx#OWLDataProperty_aaa77578_43f1_4b25_9e75_c8fdd0126c56
102 :OWLDataProperty_aaa77578_43f1_4b25_9e75_c8fdd0126c56 rdfs:type owl:DatatypeProperty ;
103         rdfs:range xsd:string ;
104         rdfs:label "has name"@en-gb .
105
106 #####
107 # Classes
108 #####
109 #####
110 #####
111 #####
112 #####
113 ↪ https://raw.githubusercontent.com/AILab-FOI/MAGO/refs/heads/main/Deliverables/Phase%202/Implementation/MAGO-Ga.owx#OWLClass_0613fb28_22ee_4c72_939f_4344f65624d7
114 :OWLClass_0613fb28_22ee_4c72_939f_4344f65624d7 rdfs:type owl:Class ;
115         rdfs:label "Feature weight"@en-gb .
116
117 #####
118 ↪ https://raw.githubusercontent.com/AILab-FOI/MAGO/refs/heads/main/Deliverables/Phase%202/Implementation/MAGO-Ga.owx#OWLClass_161fdab5_a39e_42b4_9cee_84a03f52ee88
119 :OWLClass_161fdab5_a39e_42b4_9cee_84a03f52ee88 rdfs:type owl:Class ;
120         rdfs:label "Personality model"@en-gb .
121
122 #####
123 ↪ https://raw.githubusercontent.com/AILab-FOI/MAGO/refs/heads/main/Deliverables/Phase%202/Implementation/MAGO-Ga.owx#OWLClass_19a73670_711c_40c7_a371_230c83aaa5b5
124 :OWLClass_19a73670_711c_40c7_a371_230c83aaa5b5 rdfs:type owl:Class ;
125         rdfs:subClassOf <http://purl.org/net/VideoGameOntology#Feature> ;
126         rdfs:label "Personality profile"@en-gb .
127
128 #####
129 ↪ https://raw.githubusercontent.com/AILab-FOI/MAGO/refs/heads/main/Deliverables/Phase%202/Implementation/MAGO-Ga.owx#OWLClass_420f7e1b_f0c8_4d4c_abfb_365f50700f53
130 :OWLClass_420f7e1b_f0c8_4d4c_abfb_365f50700f53 rdfs:type owl:Class ;
131         rdfs:subClassOf :OWLClass_b3f8233f_1eca_4632_895f_8ec633cf37e4 ;
132         rdfs:label "Facet"@en-gb .
133
134 #####
135 ↪ https://raw.githubusercontent.com/AILab-FOI/MAGO/refs/heads/main/Deliverables/Phase%202/Implementation/MAGO-Ga.owx#OWLClass_5f4dc944_3afc_4327_873a_483885f4114b
136 :OWLClass_5f4dc944_3afc_4327_873a_483885f4114b rdfs:type owl:Class ;
137         rdfs:subClassOf <http://purl.org/net/VideoGameOntology#Feature> ;
138         rdfs:label "Gender"@en-gb .
139
140 #####
141 ↪ https://raw.githubusercontent.com/AILab-FOI/MAGO/refs/heads/main/Deliverables/Phase%202/Implementation/MAGO-Ga.owx#OWLClass_6dae9e4d_d430_445c_b054_d2db632a0da9
142 :OWLClass_6dae9e4d_d430_445c_b054_d2db632a0da9 rdfs:type owl:Class ;
143         rdfs:subClassOf
144         ↪ <http://personales.upv.es/ccarrasc/ooooaflsmas#Objective> ;
145         rdfs:label "Gamification goal"@en-gb .
146
147 #####
148 ↪ https://raw.githubusercontent.com/AILab-FOI/MAGO/refs/heads/main/Deliverables/Phase%202/Implementation/MAGO-Ga.owx#OWLClass_abf22d34_f468_41bb_826c_c555a1846b3f
149 :OWLClass_abf22d34_f468_41bb_826c_c555a1846b3f rdfs:type owl:Class ;
150         rdfs:subClassOf :OWLClass_b3f8233f_1eca_4632_895f_8ec633cf37e4 ;
151         rdfs:label "Trait"@en-gb .
152
153 #####
154 ↪ https://raw.githubusercontent.com/AILab-FOI/MAGO/refs/heads/main/Deliverables/Phase%202/Implementation/MAGO-Ga.owx#OWLClass_ade1dd36_ed64_46ec_842b_e97727baed7f
155 :OWLClass_ade1dd36_ed64_46ec_842b_e97727baed7f rdfs:type owl:Class ;
156         rdfs:subClassOf <http://purl.org/net/VideoGameOntology#Feature> ;
157         rdfs:label "Age"@en-gb .

```

```

148 ###
149 ↪ https://raw.githubusercontent.com/AILab-FOI/MAGO/refs/heads/main/Deliverables/Phase%202/Implementation/MAGO-Ga.owx#OWLClass_b3f8233f_1eca_4632_895f_8ec633cf37e4
150 :OWLClass_b3f8233f_1eca_4632_895f_8ec633cf37e4 rdf:type owl:Class ;
151 rdfs:label "Personality feature"@en-gb .
152
153 ###
154 ↪ https://raw.githubusercontent.com/AILab-FOI/MAGO/refs/heads/main/Deliverables/Phase%202/Implementation/MAGO-Ga.owx#OWLClass_d7900a7c_c25b_4ea3_ad4e_8716bbf24045
155 :OWLClass_d7900a7c_c25b_4ea3_ad4e_8716bbf24045 rdf:type owl:Class ;
156 rdfs:subClassOf
157 ↪ <http://personales.upv.es/ccarrasc/ooooaflsmas#KnowledgeArtifact> ;
158 rdfs:label "Gamification technique"@en-gb .
159
160 #####
161 # Individuals
162 #####
163 ###
164 ↪ https://raw.githubusercontent.com/AILab-FOI/MAGO/refs/heads/main/Deliverables/Phase%202/Implementation/MAGO-Ga.owx#OWLNamedIndividual_dc90836e_f55d_4c85_93b4_10b826be7ff7
165 :OWLNamedIndividual_dc90836e_f55d_4c85_93b4_10b826be7ff7 rdf:type owl:NamedIndividual ,
166 :OWLClass_d7900a7c_c25b_4ea3_ad4e_8716bbf24045 ;
167 rdfs:label "price optimiser"@en-gb .
168
169 ### Generated by the OWL API (version 4.5.29.2024-05-13T12:11:03Z) https://github.com/owlcs/owlapi

```

Listing 3.7: Turtle serialization of the MAGO-Ga ontology

MAGO-Ga OWL/XML Serialization

```
1 <?xml version="1.0"?>
2 <Ontology xmlns="http://www.w3.org/2002/07/owl#"
3
4   <!-- xml:base="https://raw.githubusercontent.com/AI4Lab-FOI/MAGO/refs/heads/main/Deliverables/Phase%202/Implementation/MAGO-Ga.owl"
5     xmlns:rdf="http://www.w3.org/1999/02/22-rdf-syntax-ns#"
6     xmlns:xml="http://www.w3.org/XML/1998/namespace"
7     xmlns:xsd="http://www.w3.org/2001/XMLSchema#"
8     xmlns:rdfs="http://www.w3.org/2000/01/rdf-schema#"
9
10    ontology:IRI="https://raw.githubusercontent.com/AI4Lab-FOI/MAGO/refs/heads/main/Deliverables/Phase%202/Implementation/MAGO-Ga.owl">
11  <Prefix name="" IRI="http://www.semanticweb.org/bogdan/ontologies/2025/5/untitled-ontology-91"/>
12  <Prefix name="owl" IRI="http://www.w3.org/2002/07/owl#" />
13  <Prefix name="rdf" IRI="http://www.w3.org/1999/02/22-rdf-syntax-ns#" />
14  <Prefix name="xml" IRI="http://www.w3.org/XML/1998/namespace/" />
15  <Prefix name="xsd" IRI="http://www.w3.org/2001/XMLSchema#" />
16  <Prefix name="rdfs" IRI="http://www.w3.org/2000/01/rdf-schema#" />
17
18  <!-- <Import>https://raw.githubusercontent.com/AI4Lab-FOI/MAGO/refs/heads/main/Deliverables/Phase%202/Implementation/GIVEn.rdf</Import>
19  <Annotation>
20    <AnnotationProperty abbreviatedIRI="rdfs:comment"/>
21    <Literal xml:lang="en">The MAGO-Ga ontology is designed to model gamified multi-agent systems modelled as
22      <!-- intelligent virtual environments.</Literal>
23    </Annotation>
24  <Annotation>
25    <AnnotationProperty abbreviatedIRI="rdfs:label"/>
26    <Literal xml:lang="en">The Gamification of an Intelligent Virtual Environment Ontology (MAGO-Ga)</Literal>
27  </Annotation>
28  <Declaration>
29    <Class IRI="#OWLClass.0613fb28.22ee.4c72.939f.4344f65624d7"/>
30  </Declaration>
31  <Declaration>
32    <Class IRI="#OWLClass.161fdab5.a39e.42b4.9cee.84a03f52ee88"/>
33  </Declaration>
34  <Declaration>
35    <Class IRI="#OWLClass.19a73670.711c.40c7.a371.230c83aaa5b5"/>
36  </Declaration>
37  <Declaration>
38    <Class IRI="#OWLClass.420f7e1b.f0c8.4d4c.abfb.365f50700f53"/>
39  </Declaration>
40  <Declaration>
41    <Class IRI="#OWLClass.5f4dc944.3afc.4327.873a.483885f4114b"/>
42  </Declaration>
43  <Declaration>
44    <Class IRI="#OWLClass.6dae9e4d.d430.445c.b054.d2db632a0da9"/>
45  </Declaration>
46  <Declaration>
47    <Class IRI="#OWLClass.abf22d34.f468.41bb.826c.c555a1846b3f"/>
48  </Declaration>
49  <Declaration>
50    <Class IRI="#OWLClass.adeldd36.ed64.46ec.842b.e97727baed7f"/>
51  </Declaration>
52  <Declaration>
53    <Class IRI="#OWLClass.b3f8233f.leca.4632.895f.8ec633cf37e4"/>
54  </Declaration>
55  <Declaration>
56    <Class IRI="#OWLClass.d7900a7c.c25b.4ea3.ad4e.8716bbf24045"/>
57  </Declaration>
58  <Declaration>
59    <ObjectProperty IRI="#OWLObjectProperty.033c65dc.4b0b.49a0.9bde.461467eb76d6"/>
60  </Declaration>
61  <Declaration>
62    <ObjectProperty IRI="#OWLObjectProperty.268d7ca3.6f86.4e57.9627.e0320f1d33d1"/>
63  </Declaration>
64  <Declaration>
65    <ObjectProperty IRI="#OWLObjectProperty.37fa3a36.2699.4e83.b0d7.8b8e6349fd73"/>
66  </Declaration>
67  <Declaration>
68    <ObjectProperty IRI="#OWLObjectProperty.67cb4fc7.8aa5.4668.a353.308e7d7b6dd1"/>
69  </Declaration>
70  <Declaration>
71    <ObjectProperty IRI="#OWLObjectProperty.68e185c9.f82f.4507.8f25.4d50e8b3cc3c"/>
72  </Declaration>
73  <Declaration>
74    <ObjectProperty IRI="#OWLObjectProperty.8c046b10.bc4c.4cb2.aa5f.98113094df12"/>
75  </Declaration>
76  <Declaration>
77    <ObjectProperty IRI="#OWLObjectProperty.a5d640f9.c2fc.479c.ab6a.0d354f65e652"/>
78  </Declaration>
79  <Declaration>
80    <ObjectProperty IRI="#OWLObjectProperty.dcd17b13.165e.4453.a7f5.24ded1c0bc7e"/>
81  </Declaration>
82  <Declaration>
83    <DataProperty IRI="#OWLObjectProperty.48390f82.26b7.4600.a674.c779002bf747"/>
84  </Declaration>
85  <Declaration>
86    <DataProperty IRI="#OWLObjectProperty.aaa77578.43f1.4b25.9e75.c8fdd0126c56"/>
87  </Declaration>
88  <Declaration>
89    <NamedIndividual IRI="#OWLNamedIndividual.dc90836e.f55d.4c85.93b4.10b826be7ff7"/>
90  </Declaration>
91  <SubClassOf>
92    <Class IRI="#OWLClass.19a73670.711c.40c7.a371.230c83aaa5b5"/>
93    <Class IRI="http://purl.org/net/VideoGameOntology#Feature"/>
94  </SubClassOf>
```

```

91 <SubClassOf>
92 <Class IRI="#OWLClass_420f7e1b_f0c8_4d4c_abfb_365f50700f53"/>
93 <Class IRI="#OWLClass_b3f8233f_1eca_4632_895f_8ec633cf37e4"/>
94 </SubClassOf>
95 <SubClassOf>
96 <Class IRI="#OWLClass_5f4dc944_3afc_4327_873a_483885f4114b"/>
97 <Class IRI="http://purl.org/net/VideoGameOntology#Feature"/>
98 </SubClassOf>
99 <SubClassOf>
100 <Class IRI="#OWLClass_6dae9e4d_d430_445c_b054_d2db632a0da9"/>
101 <Class IRI="http://personales.upv.es/ccarrasc/ooooflsmas#Objective"/>
102 </SubClassOf>
103 <SubClassOf>
104 <Class IRI="#OWLClass_abf22d34_f468_41bb_826c_c555a1846b3f"/>
105 <Class IRI="#OWLClass_b3f8233f_1eca_4632_895f_8ec633cf37e4"/>
106 </SubClassOf>
107 <SubClassOf>
108 <Class IRI="#OWLClass_ade1dd36_ed64_46ec_842b_e97727baed7f"/>
109 <Class IRI="http://purl.org/net/VideoGameOntology#Feature"/>
110 </SubClassOf>
111 <SubClassOf>
112 <Class IRI="#OWLClass_d7900a7c_c25b_4ea3_ad4e_8716bbf24045"/>
113 <Class IRI="http://personales.upv.es/ccarrasc/ooooflsmas#KnowledgeArtifact"/>
114 </SubClassOf>
115 <ClassAssertion>
116 <Class IRI="#OWLClass_d7900a7c_c25b_4ea3_ad4e_8716bbf24045"/>
117 <NamedIndividual IRI="#OWLNamedIndividual_dc90836e_f55d_4c85_93b4_10b826be7ff7"/>
118 </ClassAssertion>
119 <InverseObjectProperties>
120 <ObjectProperty IRI="#OWLObjectProperty_67cb4fc7_8aa5_4668_a353_308e7d7b6dd1"/>
121 <ObjectProperty IRI="#OWLObjectProperty_68e185c9_f82f_4507_8f25_4d50e8b3cc3c"/>
122 </InverseObjectProperties>
123 <FunctionalObjectProperty>
124 <ObjectProperty IRI="#OWLObjectProperty_37fa3a36_2699_4e83_b0d7_8b8e6349fd73"/>
125 </FunctionalObjectProperty>
126 <FunctionalObjectProperty>
127 <ObjectProperty IRI="#OWLObjectProperty_a5d640f9_c2fc_479c_ab6a_0d354f65e652"/>
128 </FunctionalObjectProperty>
129 <ObjectPropertyDomain>
130 <ObjectProperty IRI="#OWLObjectProperty_033c65dc_4b0b_49a0_9bde_461467eb76d6"/>
131 <Class IRI="#OWLClass_d7900a7c_c25b_4ea3_ad4e_8716bbf24045"/>
132 </ObjectPropertyDomain>
133 <ObjectPropertyDomain>
134 <ObjectProperty IRI="#OWLObjectProperty_268d7ca3_6f86_4e57_9627_e0320f1d33d1"/>
135 <Class IRI="#OWLClass_d7900a7c_c25b_4ea3_ad4e_8716bbf24045"/>
136 </ObjectPropertyDomain>
137 <ObjectPropertyDomain>
138 <ObjectProperty IRI="#OWLObjectProperty_37fa3a36_2699_4e83_b0d7_8b8e6349fd73"/>
139 <Class IRI="#OWLClass_0613fb28_22ee_4c72_939f_4344f65624d7"/>
140 </ObjectPropertyDomain>
141 <ObjectPropertyDomain>
142 <ObjectProperty IRI="#OWLObjectProperty_68e185c9_f82f_4507_8f25_4d50e8b3cc3c"/>
143 <Class IRI="#OWLClass_b3f8233f_1eca_4632_895f_8ec633cf37e4"/>
144 </ObjectPropertyDomain>
145 <ObjectPropertyDomain>
146 <ObjectProperty IRI="#OWLObjectProperty_8c046b10_bc4c_4cb2_aaf8_98113094df12"/>
147 <Class IRI="http://personales.upv.es/ccarrasc/JaCaLIVE_Ontology#Observable_Event"/>
148 </ObjectPropertyDomain>
149 <ObjectPropertyDomain>
150 <ObjectProperty IRI="#OWLObjectProperty_a5d640f9_c2fc_479c_ab6a_0d354f65e652"/>
151 <ObjectUnionOf>
152 <Class IRI="http://personales.upv.es/ccarrasc/JaCaLIVE_Ontology#Agent"/>
153 <Class IRI="#OWLClass_0613fb28_22ee_4c72_939f_4344f65624d7"/>
154 <Class IRI="#OWLClass_d7900a7c_c25b_4ea3_ad4e_8716bbf24045"/>
155 </ObjectUnionOf>
156 </ObjectPropertyDomain>
157 <ObjectPropertyDomain>
158 <ObjectProperty IRI="#OWLObjectProperty_dcd17b13_165e_4453_a7f5_24ded1c8bc7e"/>
159 <Class IRI="http://personales.upv.es/ccarrasc/JaCaLIVE_Ontology#Agent"/>
160 </ObjectPropertyDomain>
161 <ObjectPropertyRange>
162 <ObjectProperty IRI="#OWLObjectProperty_033c65dc_4b0b_49a0_9bde_461467eb76d6"/>
163 <Class IRI="http://purl.org/net/VideoGameOntology#GainEvent"/>
164 </ObjectPropertyRange>
165 <ObjectPropertyRange>
166 <ObjectProperty IRI="#OWLObjectProperty_268d7ca3_6f86_4e57_9627_e0320f1d33d1"/>
167 <Class IRI="#OWLClass_6dae9e4d_d430_445c_b054_d2db632a0da9"/>
168 </ObjectPropertyRange>
169 <ObjectPropertyRange>
170 <ObjectProperty IRI="#OWLObjectProperty_37fa3a36_2699_4e83_b0d7_8b8e6349fd73"/>
171 <Class IRI="#OWLClass_b3f8233f_1eca_4632_895f_8ec633cf37e4"/>
172 </ObjectPropertyRange>
173 <ObjectPropertyRange>
174 <ObjectProperty IRI="#OWLObjectProperty_68e185c9_f82f_4507_8f25_4d50e8b3cc3c"/>
175 <Class IRI="#OWLClass_161fdab5_a39e_42b4_9cee_84a03f52ee88"/>
176 </ObjectPropertyRange>
177 <ObjectPropertyRange>
178 <ObjectProperty IRI="#OWLObjectProperty_8c046b10_bc4c_4cb2_aaf8_98113094df12"/>
179 <Class IRI="http://personales.upv.es/ccarrasc/JaCaLIVE_Ontology#Physical_Artifact"/>
180 </ObjectPropertyRange>
181 <ObjectPropertyRange>
182 <ObjectProperty IRI="#OWLObjectProperty_a5d640f9_c2fc_479c_ab6a_0d354f65e652"/>
183 <Class IRI="#OWLClass_19a73670_711c_40c7_a371_230c83aa5b5"/>
184 </ObjectPropertyRange>
185 <ObjectPropertyRange>
186 <ObjectProperty IRI="#OWLObjectProperty_dcd17b13_165e_4453_a7f5_24ded1c8bc7e"/>
187 <Class IRI="http://personales.upv.es/ccarrasc/JaCaLIVE_Ontology#Observable_Property"/>
188 </ObjectPropertyRange>
189 <FunctionalDataProperty>
190 <DataProperty IRI="#OWLDataProperty_40390f82_26b7_4600_a674_c779002bf747"/>
191 </FunctionalDataProperty>
192 <DataPropertyDomain>
193 <DataProperty IRI="#OWLDataProperty_40390f82_26b7_4600_a674_c779002bf747"/>
194 <Class IRI="#OWLClass_0613fb28_22ee_4c72_939f_4344f65624d7"/>
195 </DataPropertyDomain>
196 <DataPropertyRange>
197 <DataProperty IRI="#OWLDataProperty_40390f82_26b7_4600_a674_c779002bf747"/>
198 <Datatype abbreviatedIRI="xsd:float"/>
199 </DataPropertyRange>
200 <DataPropertyRange>
201 <DataProperty IRI="#OWLDataProperty_aaa77578_43f1_4b25_9e75_c8fdd0126c56"/>
202 <Datatype abbreviatedIRI="xsd:string"/>
203 </DataPropertyRange>
204 <AnnotationAssertion>
205 <AnnotationProperty abbreviatedIRI="rdfs:label"/>
206 <IRI>#OWLClass_0613fb28_22ee_4c72_939f_4344f65624d7</IRI>
207 <Literal xml:lang="en-gb">Feature weight</Literal>
208 </AnnotationAssertion>

```

```

209 <AnnotationAssertion>
210 <AnnotationProperty abbreviatedIRI="rdfs:label"/>
211 <IRI>#OWLClass.161fdab5_a39e_42b4_9cee_84a03f52ee88</IRI>
212 <Literal xml:lang="en-gb">Personality model</Literal>
213 </AnnotationAssertion>
214 <AnnotationAssertion>
215 <AnnotationProperty abbreviatedIRI="rdfs:label"/>
216 <IRI>#OWLClass.19a73670_711c_40c7_a371_230c83aa5b5</IRI>
217 <Literal xml:lang="en-gb">Personality profile</Literal>
218 </AnnotationAssertion>
219 <AnnotationAssertion>
220 <AnnotationProperty abbreviatedIRI="rdfs:label"/>
221 <IRI>#OWLClass.420f7e1b_f0c8_4d4c_abfb_365f50700f53</IRI>
222 <Literal xml:lang="en-gb">Facet</Literal>
223 </AnnotationAssertion>
224 <AnnotationAssertion>
225 <AnnotationProperty abbreviatedIRI="rdfs:label"/>
226 <IRI>#OWLClass.5f4dc944_3afc_4327_873a_483885f4114b</IRI>
227 <Literal xml:lang="en-gb">Gender</Literal>
228 </AnnotationAssertion>
229 <AnnotationAssertion>
230 <AnnotationProperty abbreviatedIRI="rdfs:label"/>
231 <IRI>#OWLClass.6dae9e4d_d430_445c_b054_d2db632a0da9</IRI>
232 <Literal xml:lang="en-gb">Gamification goal</Literal>
233 </AnnotationAssertion>
234 <AnnotationAssertion>
235 <AnnotationProperty abbreviatedIRI="rdfs:label"/>
236 <IRI>#OWLClass_abf22d34_f460_41bb_026c_c55a1846b3f</IRI>
237 <Literal xml:lang="en-gb">Trait</Literal>
238 </AnnotationAssertion>
239 <AnnotationAssertion>
240 <AnnotationProperty abbreviatedIRI="rdfs:label"/>
241 <IRI>#OWLClass_adeldd36_ed64_46ec_842b_e97727baed7f</IRI>
242 <Literal xml:lang="en-gb">Age</Literal>
243 </AnnotationAssertion>
244 <AnnotationAssertion>
245 <AnnotationProperty abbreviatedIRI="rdfs:label"/>
246 <IRI>#OWLClass_b3f8233f_1eca_4632_895f_0ec633cf37e4</IRI>
247 <Literal xml:lang="en-gb">Personality feature</Literal>
248 </AnnotationAssertion>
249 <AnnotationAssertion>
250 <AnnotationProperty abbreviatedIRI="rdfs:label"/>
251 <IRI>#OWLClass_d7900a7c_c25b_4ea3_ad4e_8716bbf24045</IRI>
252 <Literal xml:lang="en-gb">Gamification technique</Literal>
253 </AnnotationAssertion>
254 <AnnotationAssertion>
255 <AnnotationProperty abbreviatedIRI="rdfs:label"/>
256 <IRI>#OWLDataProperty.40390f02_26b7_4600_a674_c779002bf747</IRI>
257 <Literal xml:lang="en-gb">has weight value</Literal>
258 </AnnotationAssertion>
259 <AnnotationAssertion>
260 <AnnotationProperty abbreviatedIRI="rdfs:label"/>
261 <IRI>#OWLDataProperty.aaa77578_43f1_4b25_9e75_c8fdd0126c56</IRI>
262 <Literal xml:lang="en-gb">has name</Literal>
263 </AnnotationAssertion>
264 <AnnotationAssertion>
265 <AnnotationProperty abbreviatedIRI="rdfs:label"/>
266 <IRI>#OWLNamedIndividual_dc90836e_f55d_4c05_93b4_10b826be7ff7</IRI>
267 <Literal xml:lang="en-gb">price optimiser</Literal>
268 </AnnotationAssertion>
269 <AnnotationAssertion>
270 <AnnotationProperty abbreviatedIRI="rdfs:label"/>
271 <IRI>#OWLObjectProperty_033c65dc_4b0b_49a0_9bde_461467eb76d6</IRI>
272 <Literal xml:lang="en-gb">has gamification outcome</Literal>
273 </AnnotationAssertion>
274 <AnnotationAssertion>
275 <AnnotationProperty abbreviatedIRI="rdfs:label"/>
276 <IRI>#OWLObjectProperty_268d7ca3_6f86_4e57_9627_e0320f1d33d1</IRI>
277 <Literal xml:lang="en-gb">has gamification goal</Literal>
278 </AnnotationAssertion>
279 <AnnotationAssertion>
280 <AnnotationProperty abbreviatedIRI="rdfs:label"/>
281 <IRI>#OWLObjectProperty_37fa3a36_2699_4e83_b0d7_8b8e6349fd73</IRI>
282 <Literal xml:lang="en-gb">has personality feature</Literal>
283 </AnnotationAssertion>
284 <AnnotationAssertion>
285 <AnnotationProperty abbreviatedIRI="rdfs:label"/>
286 <IRI>#OWLObjectProperty_67cb4fc7_8aa5_4668_a353_308e7d7b6dd1</IRI>
287 <Literal xml:lang="en-gb">defines feature</Literal>
288 </AnnotationAssertion>
289 <AnnotationAssertion>
290 <AnnotationProperty abbreviatedIRI="rdfs:label"/>
291 <IRI>#OWLObjectProperty_68e185c9_f82f_4507_8f25_4d50e8b3cc3c</IRI>
292 <Literal xml:lang="en-gb">is a part of model</Literal>
293 </AnnotationAssertion>
294 <AnnotationAssertion>
295 <AnnotationProperty abbreviatedIRI="rdfs:label"/>
296 <IRI>#OWLObjectProperty_8c046b10_bc4c_4cb2_aaf8_98113094df12</IRI>
297 <Literal xml:lang="en-gb">has event object</Literal>
298 </AnnotationAssertion>
299 <AnnotationAssertion>
300 <AnnotationProperty abbreviatedIRI="rdfs:label"/>
301 <IRI>#OWLObjectProperty_a5d640f9_c2fc_479c_ab6a_0d354f65e652</IRI>
302 <Literal xml:lang="en-gb">has personality profile</Literal>
303 </AnnotationAssertion>
304 <AnnotationAssertion>
305 <AnnotationProperty abbreviatedIRI="rdfs:label"/>
306 <IRI>#OWLObjectProperty_dcd17b13_165e_4453_a7f5_24ded1c0bc7e</IRI>
307 <Literal xml:lang="en-gb">has observable property</Literal>
308 </AnnotationAssertion>
309 </Ontology>
310
311
312
313

```

<!-- Generated by the OWL API (version 4.5.29.2024-05-13T12:11:03Z) <https://github.com/owlcs/owlapi> -->

Listing 3.8: OWL/XML serialization of the MAGO-Ga ontology

Bibliography

- [1] M. Fernández-López, A. Gómez-Pérez and N. Juristo, 'METHONTOLOGY: From Ontological Art Towards Ontological Engineering,' in *Proceedings of the Ontological Engineering AAAI-97 Spring Symposium Series*, Stanford US-CA: Stanford University, 1997, pp. 33–40.
- [2] B. Okreša Đurić, 'Organizational Modeling of Large-Scale Multi-Agent Systems with Application to Computer Games,' Doctoral thesis, University of Zagreb, Varaždin HR, 2018, 236 pp.
- [3] R. Iqbal, M. A. A. Murad, A. Mustapha and N. M. Sharef, 'An Analysis of Ontology Engineering Methodologies: A Literature Review,' *Research Journal of Applied Sciences, Engineering and Technology*, vol. 6, no. 16, pp. 2993–3000, 2013, ISSN: 20407459.
- [4] B. Okreša Đurić, J. Rincon, C. Carrascosa, M. Schatten and V. Julian, 'MAMbO5: A new Ontology Approach for Modelling and Managing Intelligent Virtual Environments Based on Multi-Agent Systems,' *Journal of Ambient Intelligence and Humanized Computing*, vol. 10, no. 9, pp. 3629–3641, 2019, ISSN: 1868-5145. DOI: [10.1007/s12652-018-1089-4](https://doi.org/10.1007/s12652-018-1089-4).
- [5] J. A. Rincon, C. Carrascosa and E. Garcia, 'Developing intelligent virtual environments using MAM5 meta-model,' in *Advances in Practical Applications of Heterogeneous Multi-Agent Systems. The PAAMS Collection*, Y. Demazeau, F. Zambonelli, J. M. Corchado and J. Bajo, Eds., ser. Lecture Notes in Computer Science 8473, Cham, CH: Springer, 2014, pp. 379–382, ISBN: 9783319075501. DOI: [10.1007/978-3-319-07551-8_43](https://doi.org/10.1007/978-3-319-07551-8_43).
- [6] M. Luck and R. Aylett, 'Applying Artificial Intelligence to Virtual Reality: Intelligent Virtual Environments,' *Applied Artificial Intelligence*, vol. 14, no. 1, pp. 3–32, 2000, ISSN: 0883-9514, 1087-6545. DOI: [10.1080/088395100117142](https://doi.org/10.1080/088395100117142).
- [7] M. Uschold and M. Gruninger, 'Ontologies: Principles, Methods and Applications,' *The Knowledge Engineering Review*, vol. 11, no. 2, pp. 93–136, 1996. DOI: [10.1017/S0269888900007797](https://doi.org/10.1017/S0269888900007797).
- [8] F. Bakkerud, 'The Ontology of Game Spatiality,' *Game Studies*, vol. 23, no. 3, 12/2023, ISSN: 1604-7982.
- [9] J. T. C. Chan and W. Y. F. Yuen, 'Digital Game Ontology: Semantic Web Approach on Enhancing Game Studies,' in *2008 9th International Conference on Computer-Aided Industrial Design and Conceptual Design*, Beijing, China: IEEE, 11/2008, pp. 425–429, ISBN: 978-1-4244-3290-5. DOI: [10.1109/CAIDCD.2008.4730603](https://doi.org/10.1109/CAIDCD.2008.4730603).
- [10] R. P. De Lope and N. Medina-Medina, 'A Comprehensive Taxonomy for Serious Games,' *Journal of Educational Computing Research*, vol. 55, no. 5, pp. 629–672, 01/09/2017, ISSN: 0735-6331. DOI: [10.1177/0735633116681301](https://doi.org/10.1177/0735633116681301).
- [11] A. O. R. Franco et al., 'An Ontology for Role Playing Game,' in *Proceedings of SBGames 2018*, Foz do Iguaçu BR: Brazilian Computer Society, 2018, pp. 615–618.
- [12] L. Juel Larsen and B. Kampmann Walther, 'The Ontology of Gameplay: Toward a New Theory,' *Games and Culture*, vol. 15, no. 6, pp. 609–631, 01/09/2020, ISSN: 1555-4120. DOI: [10.1177/1555412019825929](https://doi.org/10.1177/1555412019825929).

- [13] R. Junior and F. Silva, 'Redefining the MDA Framework—The Pursuit of a Game Design Ontology,' *Information*, vol. 12, no. 10, p. 395, 10/10/2021, ISSN: 2078-2489. DOI: [10.3390/info12100395](https://doi.org/10.3390/info12100395).
- [14] J. Parkkila et al., 'An Ontology for Videogame Interoperability,' *Multimedia Tools and Applications*, vol. 76, no. 4, pp. 4981–5000, 01/02/2017, ISSN: 1573-7721. DOI: [10.1007/s11042-016-3552-6](https://doi.org/10.1007/s11042-016-3552-6).
- [15] O. R. Rocha and C. F. Zucker, 'Ludo: An Ontology to Create Linked Data Driven Serious Games,' presented at the ISWC 2015 - Workshop on LINKed EDucation, LINKED 2015, At Bethlehem, Pennsylvania, United States, 11/10/2015.
- [16] O. Sacco. 'The Core Game Ontology (CGO),' The Core Game Ontology (CGO), Accessed: 29th Oct. 2024. [Online]. Available: <https://autosemanticgame.institutedigitalgames.com/ontologies/core-game-ontology/>.
- [17] O. Sacco, A. Liapis and G. N. Yannakakis, 'Game Character Ontology (GCO): A Vocabulary for Extracting and Describing Game Character Information from Web Content,' in *Proceedings of the 13th International Conference on Semantic Systems*, Amsterdam NL: ACM, 11/09/2017, pp. 9–16, ISBN: 978-1-4503-5296-3. DOI: [10.1145/3132218.3132233](https://doi.org/10.1145/3132218.3132233).
- [18] S. Teixeira, R. V. Boas, F. Oliveira, C. Araújo and P. R. Henriques, 'OntoJogo: An Ontology for Game Classification,' in *2020 IEEE 8th International Conference on Serious Games and Applications for Health (SeGAH)*, 08/2020, pp. 1–8. DOI: [10.1109/SeGAH49190.2020.9201876](https://doi.org/10.1109/SeGAH49190.2020.9201876).
- [19] J. P. Zagal and A. Bruckman, 'The Game Ontology Project: Supporting Learning While Contributing Authentically to Game Studies,' in *Proceedings of the 8th International Conference on International Conference for the Learning Sciences - Volume 2*, ser. ICLS'o8, Utrecht, The Netherlands: International Society of the Learning Sciences, 24/06/2008, pp. 499–506.
- [20] A. O. R. Franco et al., 'An Ontology for Role Playing Games,' in *Proceedings of SBGames 2018*, Brazil, 2018, p. 4.
- [21] R. Hunicke, M. LeBlanc and R. Zubek, 'MDA: A Formal Approach to Game Design and Game Research,' in *Game Design and Tuning Workshop*, San Jose US-CA, 2004.
- [22] S. Russell and P. Norvig, *Artificial Intelligence: A Modern Approach* (Prentice Hall Series in Artificial Intelligence), 3rd ed., S. Russell and P. Norvig, red. New Jersey, USA: Prentice Hall, 2010, 1132 pp., ISBN: 978-0-13-604259-4.
- [23] J. Schell, *The Art of Game Design: A Book of Lenses*, 3rd ed. Boca Raton, US-FL: Taylor & Francis Group, 2019, ISBN: 978-1-138-63209-7.
- [24] J. Parkkila, F. Radulovic, M. Poveda and D. Garijo. 'The Video Game Ontology,' Accessed: 29th Jul. 2018. [Online]. Available: <http://vocab.linkeddata.es/vgo/>.
- [25] R. R. McCrae and P. T. Costa, *Personality in Adulthood: A Five-Factor Theory Perspective*, 2nd ed. New York US-NY: Guilford Press, 2006, 268 pp., ISBN: 978-1-57230-827-5.
- [26] S. Deterding, D. Dixon, R. Khaled and L. E. Nacke, 'Gamification: Toward a Definition,' in *Proceedings of the Gamification Workshop Proceedings*, Vancouver, CA, 2011.
- [27] R. Bouzidi, A. De Nicola, F. Nader and R. Chalal, 'OntoGamif: A modular ontology for integrated gamification,' *Applied Ontology*, vol. 14, no. 3, pp. 215–249, 19/08/2019, ISSN: 18758533, 15705838. DOI: [10.3233/A0-190212](https://doi.org/10.3233/A0-190212).
- [28] D. Dermeval, J. Albuquerque, I. I. Bittencourt, S. Isotani, A. P. Silva and J. Vassileva, 'GaTO: An Ontological Model to Apply Gamification in Intelligent Tutoring Systems,' *Frontiers in Artificial Intelligence*, vol. 2, p. 13, 31/07/2019, ISSN: 2624-8212. DOI: [10.3389/frai.2019.00013](https://doi.org/10.3389/frai.2019.00013).
- [29] G. C. Chalco, D. A. Moreira, I. I. Bittencourt, R. Mizoguchi and S. Isotani, 'Personalization of Gamification in Collaborative Learning Contexts using Ontologies,' *IEEE Latin America Transactions*, vol. 13, no. 6, pp. 1995–2002, 06/2015, ISSN: 1548-0992. DOI: [10.1109/TLA.2015.7164227](https://doi.org/10.1109/TLA.2015.7164227).

- [30] S. Bennani, A. Maalel and H. B. Ghezala, 'AGE-Learn: Ontology-Based Representation of Personalized Gamification in E-learning,' *Procedia Computer Science*, Knowledge-Based and Intelligent Information & Engineering Systems: Proceedings of the 24th International Conference KES2020, vol. 176, pp. 1005–1014, 01/01/2020, ISSN: 1877-0509. DOI: [10.1016/j.procs.2020.09.096](https://doi.org/10.1016/j.procs.2020.09.096).
- [31] S. Missaoui, S. Bennani and A. Maalel, 'Towards an Ontology for Representing a Student's Profile in Adaptive Gamified Learning System:' in *Proceedings of the 12th International Joint Conference on Knowledge Discovery, Knowledge Engineering and Knowledge Management*, Budapest HU: SCITEPRESS - Science and Technology Publications, 2020, pp. 149–156, ISBN: 978-989-758-474-9. DOI: [10.5220/0010107301490156](https://doi.org/10.5220/0010107301490156).
- [32] P. T. Palomino, A. M. Toda, L. Rodrigues, W. Oliveira, L. Nacke and S. Isotani, 'An Ontology for Modelling User' Profiles and Activities in Gamified Education,' *Research and Practice in Technology Enhanced Learning*, vol. 18, p. 018, 28/02/2023, ISSN: 1793-2068. DOI: [10.58459/rptel.2023.18018](https://doi.org/10.58459/rptel.2023.18018).
- [33] M. J. Gomez, J. A. Ruipérez-Valiente and F. J. G. Clemente, 'Developing and Validating Interoperable Ontology-Driven Game-Based Assessments,' *Expert Systems with Applications*, vol. 248, p. 123 370, 15/08/2024, ISSN: 0957-4174. DOI: [10.1016/j.eswa.2024.123370](https://doi.org/10.1016/j.eswa.2024.123370).
- [34] K. Ertan Şevk and S. Arkün Kocadere, 'A Review on Personalization of Gamified Learning Environments,' *Kuramsal Eğitimbilim*, vol. 17, no. 4, pp. 873–896, 31/10/2024, ISSN: 1308-1659. DOI: [10.30831/akukeg.1478810](https://doi.org/10.30831/akukeg.1478810).
- [35] J. Kirchner-Krath et al., 'Uncovering the Theoretical Basis of User Types: An Empirical Analysis and Critical Discussion of User Typologies in Research on Tailored Gameful Design,' *International Journal of Human-Computer Studies*, vol. 190, p. 103 314, 01/10/2024, ISSN: 1071-5819. DOI: [10.1016/j.ijhcs.2024.103314](https://doi.org/10.1016/j.ijhcs.2024.103314).
- [36] M. Phosandarack, E. Avril, S. Lepreux, L. Wallard and C. Kolski, 'User-Centered Personalized Gamification: An Umbrella Review,' *User Modeling and User-Adapted Interaction*, vol. 35, no. 1, p. 3, 20/01/2025, ISSN: 1573-1391. DOI: [10.1007/s11257-024-09423-z](https://doi.org/10.1007/s11257-024-09423-z).
- [37] A. Ray, M. Jain, L. Ma, K. H. Alhamzi, A. Ray and L. She, 'The Impact of Personality Traits, Barriers and Gamification on Gen X Continuance Intention for Mobile Credit Bill Payment Apps,' *Journal of Financial Services Marketing*, vol. 29, no. 3, pp. 1154–1174, 01/09/2024, ISSN: 1479-1846. DOI: [10.1057/s41264-023-00264-2](https://doi.org/10.1057/s41264-023-00264-2).
- [38] M. Fontana and P. Terna, 'From Agent-based models to network analysis (and return): The policy-making perspective,' University of Turin Department of Economics and Statistics "Cognetti de Martiis", Torino IT, 201507, 2015, pp. 1–19.
- [39] M. E. Gregori, J. P. Cámara and G. A. Bada, 'A jabber-based multi-agent system platform,' in *Proceedings of the Fifth International Joint Conference on Autonomous Agents and Multiagent Systems - AAMAS '06*, New York US-NY: ACM Press, 2006, p. 1282, ISBN: 1-59593-303-4. DOI: [10.1145/1160633.1160866](https://doi.org/10.1145/1160633.1160866).
- [40] J. Palanca, A. Terrasa, V. Julian and C. Carrascosa, 'SPADE 3: Supporting the New Generation of Multi-Agent Systems,' *IEEE Access*, vol. 8, pp. 182 537–182 549, 2020, ISSN: 2169-3536. DOI: [10.1109/ACCESS.2020.3027357](https://doi.org/10.1109/ACCESS.2020.3027357).
- [41] J. Palanca, *Smart Python Agent Development Environment (SPADE)*, 17/02/2025.

Notes

1. <http://www.dictionary.com/browse/goal>
2. <https://medium.com/the-mission/why-you-should-change-your-goals-into-quests-2467bbef9867>
3. define further, i.e. a main quest in a game can consist of several tasks, which can be quests themselves

